LungNet-RiskX: A Convolutional Model for Lung Cancer Detection, Nodule Categorization, and Personalized Risk Scoring

A thesis report submitted to the department of Computer Science and Engineering of the World University of Bangladesh in partial fulfillment of the requirement for award of the degree of Bachelor of Science in Computer Science & Engineering

Submitted By:

Joy Chandra Sarker	Md. Riaj	Pranto Saha
Id: wub03/22/60/3545	Id: wub03/22/60/3534	Id: wub03/22/60/3533

Supervised by:

Noshin Un Noor, Lecturer
Dept. of Computer Science and Engineering

World University of Bangladesh
Department of Computer Science and Engineering

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LETTER OF TRANSMITTAL

March 02, 2026

To

Noshin Un Noor

Lecturer

Department of Computer Science & Engineering World University of Bangladesh (WUB)

Subject: Submission of Project Report.

Dear Madam,

We are delighted to submit our project report titled **"LungNet-RiskX: A Convolutional Model for Lung Cancer Detection, Nodule Categorization, and Personalized Risk Scoring"**. It was a excellent satisfaction to work on such an valuable topic. The report is able according to the musts and guidelines of the Department of Computer Science and Engineering, World University of Bangladesh (WUB).

I consider that the report will benefit you in assessing my project work. It would be a great enjoyment for me to decipher any part or entire of the report whenever essential.

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Joy Chandra Sarker Id: wub03/22/60/3545	Md. Riaj Id: wub03/22/60/3534	Pranto Saha Id: wub03/22/60/3533

World University of Bangladesh

DECLARATION

We herewith solemnly state that the project work entitled **"LungNet-RiskX: A Convolutional Model for Lung Cancer Detection, Nodule Categorization, and Personalized Risk Scoring"**, has been supervised by Noshin Un Noor, Lecturer , designation of the department of Computer Science & Engineering, World University of Bangladesh. We ensure that the project report has not been submitted either in whole or part for any degree or Diploma in any university previously.

We hereby warrant that the work we have presented does not breach any existing copyright rule.

We additional agree to protect the university against any loss or dent arising from breach of the forgoing requirement.

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Joy Chandra Sarker Id: wub03/22/60/3545	Md. Riaj Id: wub03/22/60/3534	Pranto Saha Id: wub03/22/60/3533

Department of Computer Science and Engineering
World University of Bangladesh

CERTIFICATE

I hereby confirm that the Project Report **on "LungNet-RiskX: A Convolutional Model for Lung Cancer Detection, Nodule Categorization, and Personalized Risk Scoring"**, is a confess record of project work executed by Joy Chandra Sarker, Md. Riaj, Pranto Saha partial fulfillment of the requirements for award of the degree of the Bachelor of Science in Computer Science and Engineering from World University of Bangladesh.

The project report has been carried out under my supervision and is a recording of the bona-fide work carried out effectively by the students.

Supervisor:

……

Noshin Un Noor

Lecturer

Department of Computer Science and Engineering

World University of Bangladesh (WUB)

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Signature of Student	Signature of Student	Signature of Student
……	……	……
Joy Chandra Sarker Id: wub03/22/60/3545	Md. Riaj Id: wub03/22/60/3534	Pranto Saha Id: wub03/22/60/3533

ABSTRACT

Lung cancer is among the most common causes of mortality that are caused by the cancer and it is mainly due to untimely diagnosis and absence of clinical intervention. Even in the even radiological procedures like computed tomography (CT) scans, chest x-rays, small nodules appeared in the form of low-contrast nodules, yet they are unable to be appropriately recognized even by skilled radiologists. To do away with these diagnostic problems, this learning presents LungNet-RiskX, the hybrid form of deep learning, which is applicable to the problem of automated lung cancer, multiclassification and probability-based risk estimation. The proposed system is trained on the basis of massive medical picture 53853 lung medical pictures of the categories of normal, Benign and Malignant modules. To ensure that the dataset is standardized and improves the success of the generalization process, a formatted preprocessing pipeline, including image resizing (64x64), pixel normalization, contrast enhancement via CLAHE, median filtering to remove noise, and data enhancement, was used. The last structure consists of Convolutional Neural Networks (CNN) to detect the spatial characteristics of the residents, Vision Transformers (ViT) to learn the interrelationship between the features at both the global level and Extreme Gradient Boosting (XGBoost) to achieve the best classification with the ensembles. The experimental analysis demonstrates that the CNN + ViT + XGBoost compound model can achieve the highest performance on all the confirmed buildings, where the general correctness measured 94.21 per cent with good ROC-AUC values in all the programs, especially in malignant detection. In addition, the probability-based risk output improves the interpretability, which provides a confidence-based prediction, thus making informed clinical decisions. Overall, LungNet -RiskX is a flexible and effective decision-support platform that can be utilised in the screening of lung cancer and the prioritisation of risks in the initial stages.

Table of Contents

LETTER OF TRANSMITTAL 2
DECLARATION 3
CERTIFICATE 4
ACKNOWLEDGEMENTS 5
ABSTRACT 6
Table of Contents 7
List Of Tables 9
List Of Figures 9
List Of Abbreviations 11
Chapter 1: Introduction 12
1.1 Introduction: 12
1.2 Objectives: 13
1.3 Justification of Study: 14
1.4 Scope of Study: 15
Chapter 2: Literature Review 17
Chapter 3: Methodology 23
3.1 System Overview: 23
3.2 Dataset Description: 24
3.3. Data Preprocessing: 24
3.4. Sample Selection: 25
3.4.1 Dataset Split Distribution 26
3.4.2. Image Normalization: 27
3.4.3. Image Resizing: 27
3.4.4. Data Augmentation 28
3.4.5 Contrast Enhancement 29
3.4.6 Noise Reduction 29
3.4.7 Handling Class Imbalance 29
3.5 Model Architecture: 30
3.5.1. Input Layer and Preprocessing Integration: 31
3.5.2 Convolutional Layers: 31
3.5.3 Max Pooling Layers: 31
3.5.4 Hierarchical Feature Learning: 31
3.6.1 EfficientNet-B3 31
3.6.2 Vision Transformer (ViT) 32
3.6.3 XGBoost Classifier 32
3.6.4 Hybrid CNN–XGBoost 33

3.6.5 CNN + EfficientNetB3 34
3.6.6 NN + Vision Transformer (ViT) 35
3.6.7 CNN + Vision Transformer + XGBoost 37
3.7 Loss Function and Optimizer: 38
3.7.1 Loss Function for Deep Learning Models 38
3.8 Justification of Methodology 39
Chapter 4. Requirement Analysis, Design, and Development 40
4.1 Computational Requirements 40
4.1.1 Environment 41
Chapter 5. Result Analysis 42
5.1. Confusion Matrix 42
5.2.1 CNN + Vit + XGBoost 43
5.2.2 Confusion Matrix Summary 43
5.2.3 ROC Curve Summary 44
5.3.1 CNN + EfficientNet-B3 47
5.3.2 Confusion Matrix Summary 47
5.3.3 ROC Curve Summary 48
5.4.1 CNN + ViT 51
5.4.2 Confusion Matrix Summary 51
5.4.3 ROC Curve Summary 51
5.5.1CNN + XGBOOST 54
5.5.2 Confusion Matrix Summary 54
5.5.3 ROC Curve Summary 55
5.6 Accuracy Comparison 57
5.8 Loss and Training Analysis 58
5.9 Train Time Comparison 61
5.10 Risk Prediction and Probability Analysis 62
5.10.1 Output-Based Risk Probability Interpretation 62
5.10.2 Risk Confidence and Decision Reliability: 63
5.10.3 Clinical Risk Interpretation: 63
5.10.4 Significance of Probability-Based Risk Modeling: 64
5.10.5 Contribution of the Risk Module: 64
Chapter 6. Conclusions 65
6.1 Future Work 65
Reference: 66

List Of Tables

Table No.	Title	Page
3.2.1	Dataset Characteristics	24
3.3.1	Pre-processing Summary Table	25
3.4.1	Dataset Split Distribution	26
3.6.5	CNN + EfficientNetB3 Model Summary	35
3.6.6	CNN + Vision Transformer (ViT) Model Summary	36
3.6.7	CNN +Vision Transformer +XGBoost Model Summary	38
5.2.1	Confusion Matrix Summary	44
5.2.1	Class-wise Performance Matrix	44
5.3.1	Confusion Matrix Summary	48
5.3.1	Class-wise Performance Matrix	48
5.4.1	Confusion Matrix Summary Table	52
5.4.1	Class-wise Performance Matrix	52
5.5.1	CNN + XGBOOST Confusion Matrix	55
5.5.1	CNN + XGBOOST Class-wise Performance Matrix	55
5.6	Performance Comparison	57

List Of Figures

Figure No	Title	Page
3.1	Workflow Diagram	23
3.4	Sample lung CT images illustrating benign, extra, malignant, and normal classes.	26
3.4.1	Sample CT Images from the Dataset	26
3.4.2	Image Normalization	27
3.4.3	Original lung CT image (120 × 120). Resized lung CT image (64 × 64)	28
3.4.4	Augmentation	28
3.4.5	Previous State and After Contrast Enhancement	29
3.4.6	Original CT Image and After Median Filtering	29
3.4.7	Class Imbalance Distribution	30
3.5	Model Architecture	30
3.6.4	CNN+XGBoost Model Flowchart	33

3.6.5	CNN+EfficientnetB3 Model Flowchart	34
3.6.6	CNN+ViT+XGBoost Model Flowchart	36
3.6.7	CNN + Vit + XGBoost Model Flowchart	37
5.2.1	CNN + Vit + XGBoost Confusion Matrix and Roc Curve	43
5.2.2	CNN + EfficientNet-B3 Confusion Matrix and Roc Curve	47
5.2.3	CNN + ViT Confusion Matrix and Roc Curve	51
5.2.4	CNN + XGBOOST Confusion Matrix and Roc Curve	54
5.7	Compare Matrix	58
5.8	Loss and Training Analysis	60
5.8	Loss and Training Analysis	61
5.9	Training Time Comparison Models	62
5.10.1	Sample Risk Prediction Output for a Lung CT Image	63

List Of Abbreviations

Short Form	Full Form
AI	Artificial Intelligence
AUC	Area Under the Curve
CLAHE	Contrast Limited Adaptive Histogram Equalization
CNN	Convolutional Neural Network
CT	Computed Tomography
DL	Deep Learning
FN	False Negative
FP	False Positive
ML	Machine Learning
OVR	One-vs-Rest
ROC	Receiver Operating Characteristic
SCCE	Sparse Categorical Cross-Entropy
TN	True Negative
TP	True Positive
ViT	Vision Transformer
XGBoost	Extreme Gradient Boosting

Chapter 1: Introduction

1.1 Introduction:

It is a known fact that lung cancer is among the most widespread and even life-threatening cancerous diseases in most regions of the world, but it is an agonizing health issue for the population. One of the primary reasons that has led to the high mortality rate currently experienced in so far as this disease is often diagnosed late, since most of the cases are only diagnosed at the advanced stage of the cancer. In line with the World Health Organization (2021), lung cancer represents one of the causes of mortality of approximately 1.8 million individuals a year and that is why it becomes extremely difficult on the international level to detect tumors at their initial and curable stages. These lesions are usually tiny and cannot be visualized with the eye, and when ordinary imaging is conducted, they are difficult to detect. The opportunities for the diagnosis of a modern imaging environment have been intensified to a great extent due to the introduction of such technologies as computed tomography (CT) scan or digital radiography; however, the initial tumors can be highly specific and inconsequential nodules, which can be difficult to identify even in the case when a highly specialized radiologist works. As a result of this fact, the diagnosis of the high percentage of patients is only made when the clinical symptoms are aggravated, thus limiting treatment opportunities and diminishing the chances of survival in the long-term. (Organization, 2021)

The conventional imaging modalities still have a major role in lung cancer diagnosis, but they have various limitations, which are inherent in them. The diagnostic uncertainty has been known to be caused by the variability of the interpretative skill of radiologists, the inconsistency of the quality and resolution of the image, and the insensitivity of the abnormalities at their earlier stages. They can make it a false-positive and a false-negative, thus leading to the postponement of appropriate clinical treatment and cure. These experiences demonstrate that automated and standardized, and highly accurate diagnostic logics are desperately required to help in assisting clinicians in making consistent and reliable judgments.

The recent advances in the category of deep learning, namely, convolutional neural networks (CNNs) have enabled it to overcome these shortcomings in the medical image analysis. VGGNet, ResNet, and U-Net CNN-based networks have been shown to be shockingly proficient in acquiring hierarchical and discriminating organization in explicit applications of raw imaging data. As far as

pulmonary nodules, localization and classification of malignancy, and other procedures have been done successfully using these cases. CNN-like approaches would be an excellent answer to due to their features of operating in complicated imaging conditions. Eliminating discrepancies within the diagnoses and enhancing the quality of tracking it at an early-stage. Nevertheless, the information presented in the imaging might not be as efficient as it is supposed to be to describe a personal predisposition to lung cancer. As the experience in the field of epidemiology proved, several non-imaging risk factors such as age, smoking habits, work exposure, air pollution, and genetic predisposition, have a great role in determining the probabilities of the disease. Such clinical and demographic factors in conjunction with the imaging-based attributes, may be applicable in the creation of a more detailed and individualized risk assessment framework that can be utilized to increase the predictive quality and result in preventive medical policies. The provided study introduces a hybrid of a lung cancer diagnosis and personal risk assessment called LungNet-RiskX that will be implemented to attract attention to the diagnosis of the pulmonary cancer and determine personal risk. It is a hybrid of Convolutional Neural Networks (CNN) and Vision Transformers (ViT) to analyze the medical image more effectively, preserving more space-specific details and more information related to details in general. In addition to image analysis, the model also incorporates substantial non-imaging risk factors, including age, smoking and environmental history (that are estimated with the help of an XGBoost prediction module) (in order to obtain a risk score, which is unique). It also has a recommendation feature that converts the measured risk into a preventive recommendation. This paper will provide a scalable and realistic proof of concept framework to detect and provide decision support on early lung cancer in a single system that will centralize the screening, classification and risk assessment of the individual.

1.2 Objectives:

1. Implement a CNN-based system, which can identify lung cancer using the CT or X-ray image based on the powerful image processing methods.
2. To come up with a mechanism which is capable of being able to make an accurate line between malignant and benign lung nodules.
3. To develop a risk guess model that would be used to estimate the percentage chance of developing lung cancer based on the factors age, previous history of smoking, exposure to the environment and family history.

1.3 Justification of Study:

Lung cancer is known to be one of the top causes of cancer-related deaths in the world, especially because most patients are diagnosed late and given medical attention late. The nature of the diagnostic process, though, requires a significant contribution of manual inspection by the radiologists is largely determined by some of the tomography techniques like Computed Tomography (CT) and chest X-rays, which form a part of the initial disclosure. The distinction between benign and cancerous nodules of the complex image data is not a painless task and it may be susceptible to inter-observer variation and mistakes in diagnosis. This limitation can lead to the death of a premature malignancy or unnecessary invasive procedures on a benign status. The motivation of the present study is that the most important is the necessity to address this gap in diagnosis, which can be critical.

The most recent innovation of deep learning has radically transformed medical image analysis. Convolutional Neural Networks (CNNs) have proved to possess tremendous potentials drawing hierarchical spatial arrangements on radiological data. Nevertheless, the traditional CNN-based models directly address the local feature representation and may be unable to address the long-range contextual dependence of the complicated lung structures. To address this limitation, it is suggested that Vision Transformers (ViT) suggest interest mechanisms that generate the association of the global context of the entire image. The study incorporates CNN and ViT frameworks into a combined framework, which allows not only localizing the feature learning process but also the global contextual feature learning and contributes to the enhanced accuracy and power of pulmonary nodule detection.

Complete clinical judgment, however, does not rely on proper image-based classification. In association with the determinants that identify the risks of lung cancer, demographic and epidemiological determinants i.e. age, the history of smoking (pack-years), occupational exposure to carcinogens, ecological pollutants and domestic antiquity of cancer are also determined by the radiological results.

Such arranged risk variables that lead to poor examination of the patient are not considered in the traditional diagnostic systems.

This gap is bridged through an ensemble-based machine learning model, which is an Extreme Gradient Boosting (XGBoost) that entails integration of imaging-based malignancy likelihoods with organized patient unique risk elements. It is a mix fusion strategy that can be developed.

a personal risk score, which brings a more holistic and clinically meaningful analysis of the risk of lung cancer.

Secondly, precision medicine and preventive measure is becoming an important aspect in the contemporary medical practice. By doing so, the paper will expand the detection and classification exercise by incorporating the recommendation component that will transform the scores of the respective risks into prevention knowledge. LungNet-RiskX offers mainly personalized recommendations, risk modelling, automatic detection, and hybrid deep feature extraction in a single framework and has the potential of enhancing the sector of early diagnosis, risk stratification and proactive management of healthcare, as the decision-support system is new and scalable.

1.4 Scope of Study:

The overall objective of the research will be to design, develop and test an intelligent hybrid diagnostic tool to diagnose lung cancers and risk prediction, on an individual basis by using the deep scholarship system and ensemble machine learning systems. The study is formulated in the specific technical and practical frames to have methodological rigor and viability.

In the analysis part of the image, the publicly available image of a CT scan and a chest X-ray is used. The use of high-level preprocessing operations, counting normalization, resizing operation, noise reduction, and data increase is used to enhance the quality of images and standardize the inputs. Besides this, the concept of hybrid deep learning architecture is adopted to achieve automated nodules detection and aggregation that integrates Convolutional Neural Networks (CNNs) with Vision Transformers (ViT). They are CNN to derive hierarchical spatial features and Vision Transformer that derives the global relative relationships through the global self-attention. A mixture of such a hybrid construction provides malignancy probability scores of detected pulmonary nodules.

Along with imaging, the study defines a more structured risk prediction module that integrates imaging-detected risk of malignancy with other corresponding non-imaging risk factors that include the age of the patient, smoking history (pack-years), carcinogen occupational or environmental exposure, and family history.

of lung cancer. A single XGBoost model is then applied on these sources of heterogenous data to create the single lung cancer risk percentage. One of the benefits of using XGBoost is the predictive performance, which is increased with the help of the power of ensemble learning and gradient boosting to improve the predictive performance based on the generalization of predictability between the profiles of different patients.

Another part of its area is the creation of a recommendation engine, which will convert the scores of the individual risks into automated preventive strategies, including smoking cessation advice or proposing the screening schedule. The paper is restricted to diagnostic aids and risk assessment, and lacks treatment planning and treatment options. Although it is compared to the extensive clinical trials and a practical application where the retrospective datasets and the measurements of implementation like truth, precisionexactness recall, F1- score, and ROC-AUC are assessed, the scope is out of its reach.

Chapter 2: Literature Review

A deep learning system was constructed to automate the process of lung cancer detection to histopathological tissue images so as to assist the pathologists in cutting the work and time spent in the diagnosis process. The researchers have used the Kaggle dataset, which is very huge with over 220,000 images and have undergone resizing, normalization, and balancing of classes steps to reach the final work count of 178,000 samples. Some of the models of transfer learning that have been considered include the InceptionV3 models, but the convolutional neural network (CNN) specific to it performed the best with an accuracy of 97.09 percent and high-level recall and specificity, and precision. Such findings indicate that task specific CNN has a higher possibility of arriving at an optimal solution as compared to generic pretrained models in such a medical image classification. Still, the application value of the model is questionable since it has been applied to a single dataset; the article does not provide an adequate description of the architecture to be completely reproducible. (Asraa, 2022)

The EFFI-CNN is a deliberate convolutional neural network design, which is proposed to address the needs of both computational efficiency and accuracy of lung cancer in CT scans. The model is trained with a simplified seven-layer sequential model where it was trained on a subset of filtered images (1,400), which were found in the public LIDC-IDRI and Mendeley repositories. The investigation results indicate that the network has a test accuracy of 87.02 percent and a recall rate of 98 percent, which is quite high and proves the high ability of the network to detect positive cases of lung cancer. Compared to the previous structure adopted by the authors, the EFFI-CNN proves to have remarkable advancement in accuracy and deal-out rate. Nonetheless, the assessment is very self-referential, and the research would be improved with a more detailed comparison to the currently existing models of state of the art besides validation on large and more diverse datasets to further determine the efficiency and quality of application of the model in clinical screening. (Tulasiramu, 2019)

The latest research of 2020 indicates that the opportunity to develop the automated lung cancer screening is provided by the development of a holistic, probabilistically-grounded deep learning structure to analyze low-dose CT scans. The article combines the two steps of detecting and diagnosing in a Chinese way to introduce a new synergistic solution, where a 3D V-Net is used to operate on nodule candidate extraction and an attention-based multiple instances learning network

is used to operate on malignancy in a joint end-to-end trained manner. Its critical characteristic is strict quantification of model uncertainty in the form of Monte Carlo dropouts and deep ensembles, which provide intelligible probability scores that effectively summarize predictive confidence, which is a major facet of clinical trust and triage. It achieves the state-of-the-art with an LUNA16 on detection of 0.921 FROC score and an AUC on the Kaggle dataset on patient-level diagnosis of 0.87, and demonstrates that a loosely-coupled CADe/CADx framework does not just perform to the state of the art but in fact improves the overall performance of the diagnostic itself by naturally learning to accept the false positives of detection. (Ozdemir, 2020)

Lung cancer is a death sentence, and the sooner it is diagnosed, it has a long way to save the lives of the victims. The paper will suggest a machine learning identification of the images using machine learning and image processing that will minimize the human error in the diagnosis process. The images obtained by the CT scan are then subjected to grayscale conversion, histogram equalization, thresholding, subdivision and figure extraction (area, perimeter, eccentricity) which are plotted using categorization by the C4.5 decision tree process. The data was 60 CT scans of 61 patients; one in every 50 patients was considered a training image. The suggested system was correct in 78 percent in the classification of the malignant and benign tumors. It was established that image processing coupled with machine learning is better in correlation to speed of detection and accuracy. Nonetheless, the research gaps follow the format of small size, the absence of multi-class classification, and external validation. The results indicate the capability of C4.5 in diagnosing the binary lung cancer. The high quality of the pre-processing being dependent, and also the paucity of the diversity of the data, is taken to be a weak point. The research finds that the cause of early diagnosis can be furthered with the help of automated frameworks, but more diverse and larger datasets are needed, and they will be more reliable. (Raut1, 2021)

The weaknesses and strengths of the current approaches were applied in designing a CAD framework, which applied a rule-based and image-processing-driven system in assisting in the discovery of lung cancer in CT scan images. The authors have examined part of the past system and they have found that watershed segmentation was the best foundation. Then, they have offered a better pipeline by first doing median and Gaussian filter to eliminate salt and pepper and then marker control watershed segmentation which identifies nodules more accurately. Other geometrical and other measures of intensity like centroid, distance, eccentricity, perimeter, and

mean pixel intensity have been derived and used to train an SVM to be able to differentiate between benign and malignant lesion which is a weakness of the previous models. The enhanced technique with the LIDC data of CT was able to improve the nodule-detection accuracy to 92 per cent and benign malignancy accuracy was 86.6 per cent. Although the method is superior to the last most effective model and it introduces an enormous diagnostic use, the outcomes are not as robust as the clinical ones, and the system fails to categorize the stages of cancer, which the authors consider the research of the future. (Makajua, 2017)

In the field of medical image analysis, to study the effort of knowledge transportation, one study in 2024 compared various popular pre-trained convolutional neural networks to the task of lung CT scan classification. Architectures, such as VGG-16, ResNet-50, Exception, Mobile Net, were also maximised on architectures using LIDC-IDRI which was also filtered (mean and anisotropic diffusion) to improve performance. It was also determined that a trained VGG-16 model with some adjustments was the most appropriate in terms of the deep learning methods with an accuracy of 86.71 to differentiate the normal, cancer and the doubtful cases. Interestingly, even the classical machine learning models, such as the SVM and KNN, proved to be ideal or close to it in their application on already obtained numerical metadata and this indicates the importance of feature representation in model accuracy. The paper ends with the realization of the possibility of ready-cooked CNNs in identifying cancer on the lung and the problems of data size and character engineering as a method of reaching better results in future studies. (Chiet1, 2024)

The systematic review creates an overview of the current developments and topicality of the deep learning (DL) applications to automated lung cancer diagnosis using computed tomography (CT) scans applied to the analysis of 80 lessons out of the initially 1,448 publications published from 2015 to 2024. This review is a clear description of a decision process of the diagnosis pipeline, i.e., detection, segmentation, and classification, and suggests that convolutional neural networks (CNNs) are the most widespread with a detector sensitivity of the highest level, 98.7% and a classification accuracy of the highest level, 99.6%. The further potential of nodule localization and the enhanced possibility to capture global contextual information can also be demonstrated with the help of hybrid architectures and the new usage of Vision Transformers (ViTs). Nevertheless, despite these encouraging findings, the review notes that a massive disconnect exists between the research performance results and clinical practical application that has been attributed to overfitting and low external validity to other groups and imaging modalities due to the overreliance

on a limited set of publicly available datasets (e.g., LIDC-IDRI and LUNA16). The other ones would be the absence of model explainability, which leads to diminished clinician confidence and excessive emphasis on binary classification tasks, which tend to ignore clinically relevant goals and objectives like cancer staging and subtyping. The conclusion of the review recommends that future studies should be focused on explainable AI (XAI), multi-centre validation on a systematic foundation across a wide variety of data sets, and creation of a standardized and multi-modal data archive so as to guarantee a successful transfer of deep learning models to an effective clinical diagnostic model. (Abdullahi, 2025)

The article contrasts the validity of a commercially offered deep learning (DL) algorithm that recognizes lung cancer on chest X-rays, hence, in the setting of a mass scale, asymptomatic health screening group, where the rate of positive disease diagnoses is horrendously low. The results reveal that the algorithm has the same diagnostic ability as the big-time radiologists. In a controlled validation test, the algorithm has accomplished an excellent area under the curve (AUC) of 0.99 and all detected 90 percent of visible lung cancers without a statistically significant difference from the radiologist panel of radiologists, with a very high rate of false positives. The sensitivity was programmed to respond to a similar percentage of 70, which was to reflect the specificity of the radiologists who were using it. The sensitivity of the algorithm also correlated directly with tumor conspicuity, as the algorithm was able to identify 83 percent of all visible cancers and 100 percent of all visible cancers in a larger set of radiographs (more than 100,000). One of the greatest additions of this work is the emphasis on realistic and low prevalence scenario of screening and expansion of the artificial constriction of the disease-enriched data used in earlier studies. The biggest methodological error, though, is the fact that they applied the long-term follow-up of the patients rather than the concurrent CT scans to establish the condition of the cancer, which can result in the neglect of the very slow-growing tumours. The outcomes show that the algorithm could have good reliability and high sensitivity when used as a first-line screening tool to aid radiologists and can help to reduce diagnostic oversight, though its ultimate usefulness as an assistant during clinical practice remains the subject of future research. (Lee, 2020)

The timely and precise differentiation of benign and malignant nodules in the lungs has been among the biggest discrepancies in the CT-based diagnosis and the given paper addresses this gap by deploying an enhanced deep-learning model. The authors present the idea of DCS web a two-

branch architecture that consists of the combination of CNN-based local feature extraction with the global contextual modeling of Swin Transformer, which is further enhanced with the help of a Conv-MLP block that promotes the interaction of the windows. Using a constant preprocessing, ten-fold cross-validation model yields an apparent performance improvement on both LUNA16 and LUNA16-K data, with the model achieving a 90.96 percent accuracy, 90.56 percent recall, and an accuracy unit curve of 0.94, surpassing other popular CNN and Transformer and hybrid models, including ResNet 50, Swin-T, FiT, Gvt, and Spiking Reformers. Although the system is highly sensitive and selective in the detection of cancer at its early phases, the authors affirm that the computation requirements and the use of the pretrained datasets can be a challenge towards the generalization of the model. Overall, the DCS web provides an effective and powerful framework of enhancing the regularity of the categorization of lung nodules in clinical practice. (Faizi1, 2025)

The ongoing project research will problematize five versions of the YOLO family of object-detection, including: YOLOv5, YOLOv8, YOLOv9, YOLOv10, and YOLOv11 on the large-scale Lung-PET-CT-Dx dataset as an attempt to determine the issue of effective localization and classification of lesions associated with lung cancer in the CT scans. Once the DICOM images were converted to the standardized images of JPG and all the models were preprocessed similarly, each was trained to identify four fundamental types of lung cancer, and the performance of each in the loss measures, mAPs, and precision and recall was compared. The more stable version was YOLOv8, which had a precision of 90.32 and a recall of 84.91 and the highest mAP50 and mAP50-95 scores, implying that it obviously had an upper hand in both localization of lesions and the classification of the type of lesions. Though the method has good diagnostic and good detection capabilities, the authors note that it has large limitations in terms of computational requirements, lacks PET/CT data and sensitivity to changes in datasets. On the whole, the paper has revealed that the YOLOv8 can be the most promising in the practical implementation of the computer-aided lung cancer detection and the role of deep learning in the enhancement of the process of early diagnostics. (Kuo-Yang Huang 1, 2025)

Early identification of the abnormality of the lungs using the chest radiographs is still a challenge, particularly in cases where the small spots of congestion are not detected during the normal clinical check-up. The work was also intended to produce the AI-based model able to make the quality of

the diagnosis better by correlating the radiograph features with the real clinical results, and not what the radiologists put on the label. It was achieved through the training of a deep-learning model on a large number of X-rays of hospital patients and their respective patient records, which subsequently made the system learn the correlation of specific patterns with known diagnoses. The accuracy of the model was 96 percent in total, which is very high compared to the application of the traditional image-only method. The key findings include the fact that outcome-focused learning helps the system to detect weak or vague abnormalities that to human beings are otherwise not noted. A dependence on retrospective data, a mixed image quality and absence of multi-center luxuries is one of the limitations, casting doubt on the applicability to a larger group of the population. Generally, it can be said that the article has a promising quality of AI to offer a more plausible and clinically valuable aid in the day-to-day interpretation of lung diseases. (Meeradevi1, 2025)

CT scanning can only be used to identify lung cancer at an early stage and even then, it becomes difficult, particularly when the datasets available are tiny and of an irregular appearance. The main aim of the research work was to find a lightweight deep learning model that would not only detect lung abnormalities with greater accuracy but also be computationally efficient. It was processed through preprocessing CT images, which was then complemented by differential augmentation to make the data more heterogeneous, and then a small convolutional neural network that would not overfit was trained. It was compared to bigger architectures such as DenseNet121, ResNet152V2 and EfficientNetB0. The precision of the proposed system was found to be 96.50 and it is clear that the proposed system is clearly better than the benchmark models as it consumes less computational resources. Notable findings demonstrate that with minimal training data, balanced feature learning is very effective with the application of differential augmentation. The weaknesses include the fact that a comparatively small dataset is involved, external multi-center validation is not carried out, and there is no information on whether the model can be applied in other imaging scenarios. On the whole, the research has demonstrated encouraging opportunities in more effective AI-based classification of lung cancer. (Shariff1, 2025)

Chapter 3: Methodology

3.1 System Overview:

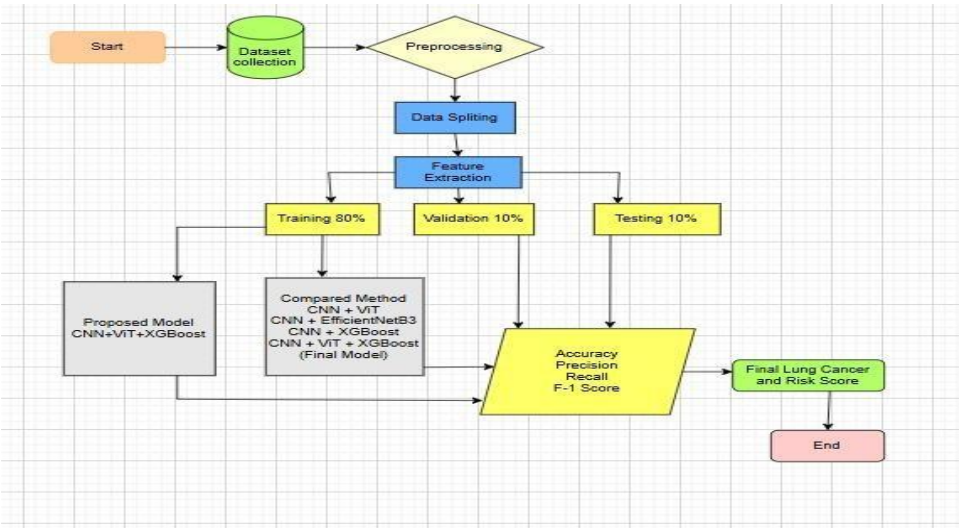

Figure 3.1: Workflow Diagram

Our dataset contained medical lung images (CT and chest X- ray images) of 53853. The data set is of three categories: the normal (healthy lungs), Benign nodules, and the Malignant tumors, which are essential in developing an effective system of lung cancer revelation and classification. Images were model-trained and pre-organized as is recommended in the proposed structure. Firstly, some pre-processing steps were undertaken, including image resizing, color pixel normalization, and data strengthening to facilitate image quality and enhance the extent of generalization of images in the model. In addition, the contrast enhancement of the lungs on CLAHE and the removal of noise on the median filter were used to reveal fine abnormalities in the lungs and prevent unwanted noise, respectively.

A second hand of the processed data was then second-hand to Pullman multiple deep knowledge models, including cross models, which syndicate Convolutional Neural Networks (CNNs) and more complex training models like Vision Transformers (CNN plus ViT) and ensemble-based learning using XGBoost. The models can be used to classify lung disease and to reliably determine the lung cancer detection capability.

3.2 Dataset Description:

We used a dataset of medical images of lungs, comprising healthy lungs, benign growths and cancerous tumors, that offered a variety to test and train our models. Its data was being broken down into 80% teaching, 10% confirmation, and 10% assessment as shown under Table 1.

Table 3.2.1: Dataset Characteristics

Dataset Characteristics	Description
Number of Images	53853
Training Set (80%)	43082 images
Test Set (10%)	5386 images
Validation(10%)	5385
Categories	Normal, Benign, Malignant
Image Format	JPEG / PNG
Color Format	RGB / Grayscale
Resolution	Resized to uniform dimensions during preprocessing

3.3. Data Preprocessing:

Data preparation is of utmost importance to our lung cancer detection scheme as medical CTs may be anywhere with brightness, sharpness, and fuzziness. Thus, before we start training our CNN and Vision Transformer model, we have used a handful of tricks to make sure that the images we feed it are not unreliable and inconsistent. All these actions are meant to make the pictures clearer and also all of the same size and also to make the model more skillful in choosing the valuable stuff.

Table 3.3.1 Pre-processing Summary Table

Technique	Purpose
Image Resizing	Uniform input size
RGB Conversion	Model compatibility
Pixel Normalization	Stable training
Dataset Splitting	Fair evaluation
Filename-based Labeling	Automated annotation
Data Augmentation	Reduce overfitting
Batch Loading	Efficient training
Shuffling	Better generalization
CLAHE	Improve image contrast for clearer nodules

3.4. Sample Selection:

Figure 3.4. represents the samples of benign, malignant and normal classes of lung CT images extracted from the dataset. The samples were exceedingly chosen in terms of balancing the classes to minimize the probability of biasness of the normal coaching practice.

In order to get a good and effective estimation of the performance, the dataset was the most effective to be differentiated into exercising, validation and challenging subsets. The total images were allocated approximately 80 percent in guidance, and the other 20 percent was to be allocated in scanning. Equally, the validation subset, which was as large as the test set was formed out of the training data. This confirmation set was utilized to run the model in the condition of exercise as well as to control risk of overfitting, which is outlined in Table 3.

The additional illustration of CT scans labelling by numerical numbers (e.g., y = 0, y <) is presented in Figure 3.4.1. Where = 1 and Y = 2 are respectively categorical classes that they are members of in supervised learning. These samples show some obvious differences in the texture patterns, contrasts, and structural characteristics of the classes. Different problems were encountered when preparing the datasets, including an imbalance in the classes, and varying image resolution, and variability in the quality of the image. All the images were subjected to standardized preprocessing schemes so as to ascertain data consistency and reliability like resizing and normalization of data. Such measures ensured consistency across the set of data and encouraged consistent model training. The discussion of the preprocessing methodology used is provided in the next section in detail.

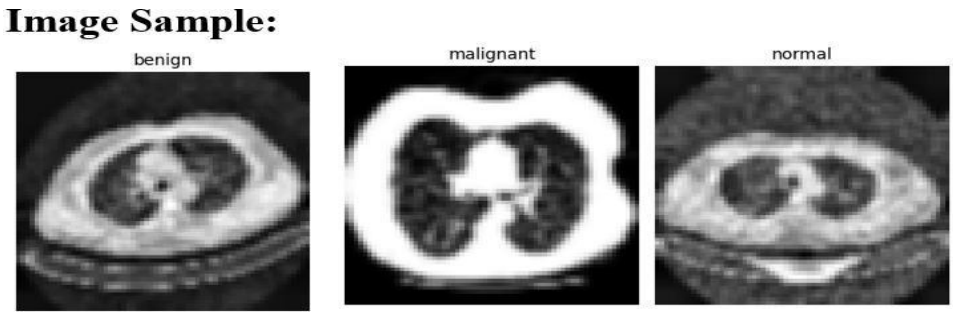

Figure 3.4. Sample lung CT images illustrating benign, extra, malignant, and normal classes.

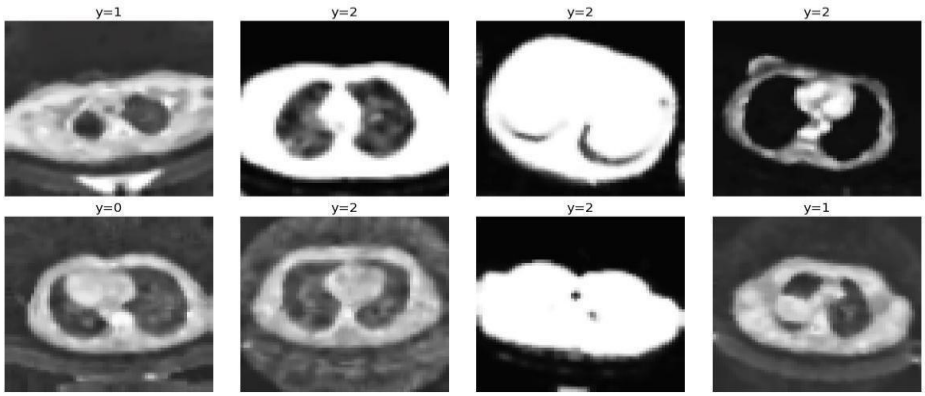

Figure 3.4.1. Sample CT Images from the Dataset

3.4.1 Dataset Split Distribution

Table 3.4.1 indicates a distribution of the dataset used in the current study. There were 43,082 (80) images in the work-out set, 5,385 (10) in the confirmation set, and 5,386 (10) in the test set out of 53,853 images. This division strategy provides adequate information to achieve effective model training and retain separate confirmation and test subsets to guarantee a sound evaluation of performance and objective evaluation.

Table 3.4.1: Dataset Split Distribution

Dataset Split	Number of Images	Percentage (%)
Training Set	43,082	≈ 80%
Validation Set	5,385	≈ 10%
Test Set	5,386	≈ 10%
Total	53,853	100%

3.4.2. Image Normalization:

Image normalization was done to minimize the intensity disparity that is prevalent across lung computer tomography (CT) images and provide numerical consistency in training the model. As a result of the variance in imaging devices, imaging strategies, and non-uniform pixel intensities are common in patient-specific conditions, where the CT images are not processed. In order to overcome this, the pixel strength values of each image were harmonized to a given limit then they were fed into the model. The advantage of such a normalization process is that the contrast and the luminosity of the whole dataset is homogeneous and, therefore, it enhances the efficacy of feature extraction. Consequently, the hybrid convolutional neural network-Vision Transformer (ViT) model is more effective in classification and has a higher convergence rate.

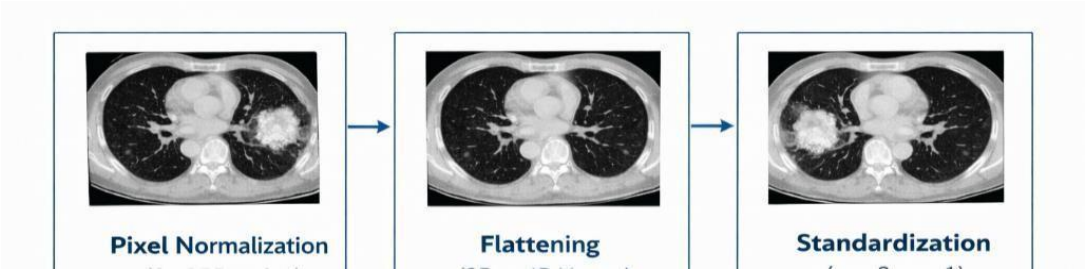

Figure 3.4.2-Image Normalization

3.4.3. Image Resizing:

The image resolution of the lung CT scans in the dataset was not homogenous that can result in instability in training. In order to check this a resizing strategy was employed where all the pictures were resized to a similar size. Specifically, each of the CT images was down-sampled to 64 x 64 pixels to guarantee that the model can be applied to the architectural model as intended with the hybrid CNN + ViT.

Whereas Figure 2 depicts an original CT image in its original resolution of 64 x 64 pixels, Figure 3 depicts the transformed image in 64 x 64 pixels. Such standardization enables representation of features in a similar manner and less computation load, hence leading to efficiency of the system in general.

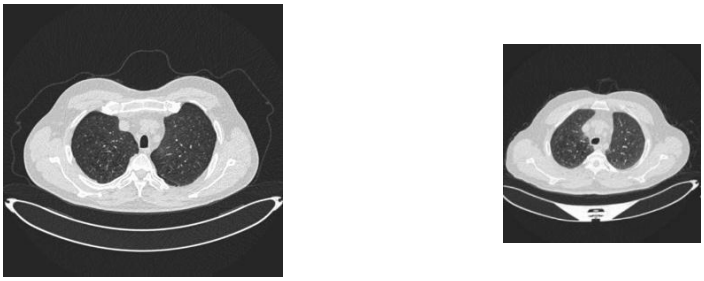

Figure 3.4.3. Original lung CT image (120 × 120). Resized lung CT image (64 × 64)

3.4.4. Data Augmentation

To enhance the diversity of the information and generalization capacity of the desired standard, data expansion systems were used on the guiding set. Because medical image data are often typified by small sample sizes, which is one of the conditions that are likely to cause overfitting in deep learning models, one of the available methods that can be applied to create new training samples, though the original class labels are preserved. Other functions were utilized like rotation, straight and upright spinning, scaling and slight alteration of amounts.

These variations work well in reproducing the natural variability in the real-life CT scans, thus, reducing the model's resistance to variations based on space and orientation. (Nyathi, 2025)

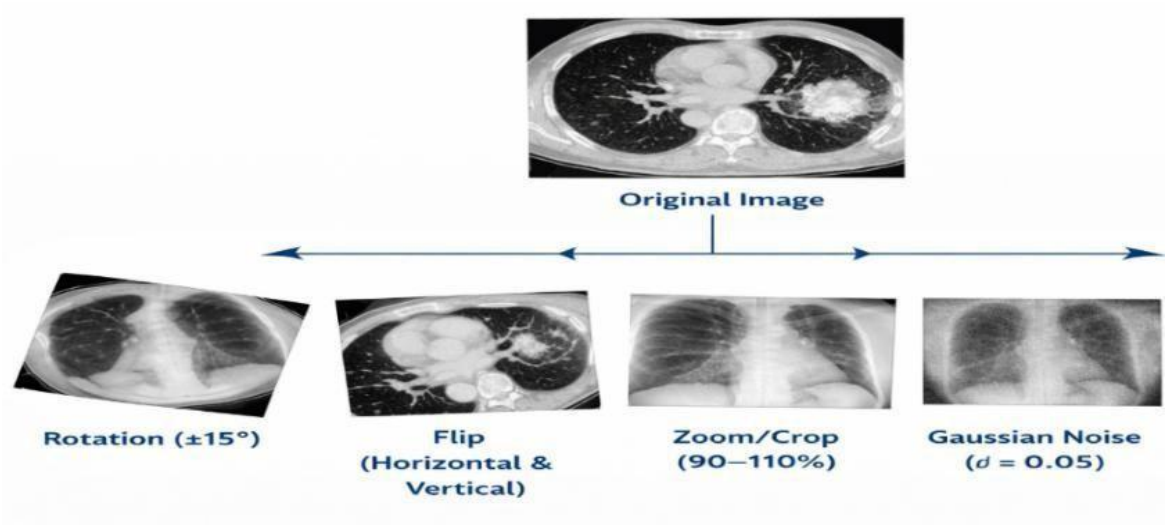

Figure 3.4.4 Augmentation

3.4.5 Contrast Enhancement

In order to increase regional differentiation and increase even more subtle texture details, relevant in determining the abnormalities at the early stages during the lung images, Contrast Limited Adaptive Histogram Equalization (CLAHE) was ordered.

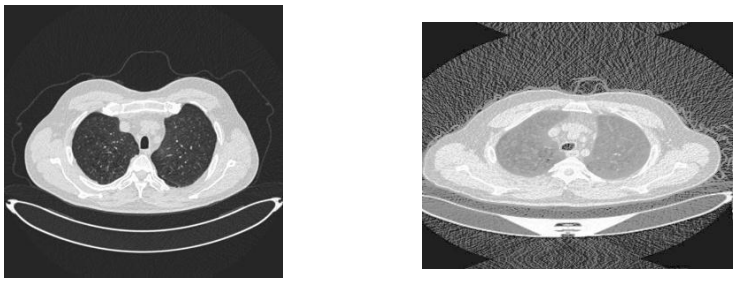

Figure: 3.4.5: Previous State and After Contrast Enhancement

3.4.6 Noise Reduction

A median filtering algorithm employing a 3x3 kernel was employed to reduce the impulse noise that is typically present in CT images, but still provide edges in the image that contain diagnostically important information. Median cleaning is extremely helpful in the salt-and-pepper cleaning and maintenance of lung boundaries.

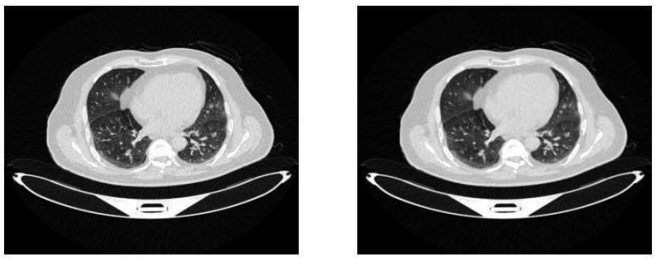

Figure: 3.4.6 Original CT Image and After Median Filtering

3.4.7 Handling Class Imbalance

The data indicates that the balance of classes is largely imbalanced, particularly regarding normal and pathological cases. To deal with this issue, we employed the balanced sampling and weighting of classes. Class increment assigns additional penalties on the subgroup classes that are to obtain equal treatment during the teaching process in comparison with balanced sampling, which ensures that all classes are equally represented in every training batch.

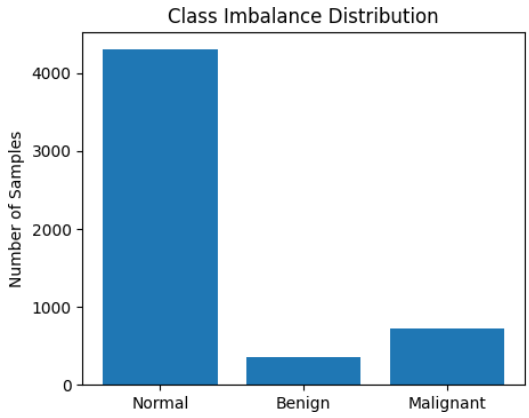

Figure: 3.4.7 Class Imbalance Distribution

3.5 Model Architecture:

The suggested lung cancer recognition framework is the application of a mixture of deep learning frameworks as a combination of Convolutional Neural Networks (CNN) and Vision Transformers (ViT). The model is aimed to process the images of the lung CTs at the standard resolution of 64 x 64 pixels and alleviate the impact of computational overhead, yet retain the essential diagnostic characteristics. The architecture is based on the principle of exploiting CNNs strengths on local spatial characteristics and self-attention mechanisms of ViTs to absorb longer range contextual dependences, which is a combination required to process lung cancer with high accuracy. (Bushara, 2022)

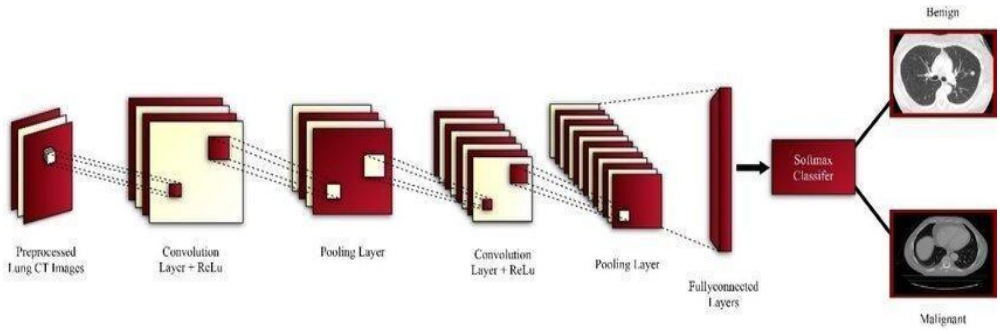

Figure: 3.5 Model Architecture

3.5.1. Input Layer and Preprocessing Integration:

The preprocessed CT images of the lung are fed into the input layer by sorting the pixel count number of 0 to 1 by resizing the lung CT images to 64 x 64 pixels and regularizing to 64 x 64 pixels. Reducing the model size to a smaller resolution results in far fewer memories used and training time, as well as makes the model computationally efficient with no reduction in classification accuracy.

3.5.2 Convolutional Layers:

A number of Conv2D filters are manipulated in order to obtain spatial characteristics of introduced pictures. Downsampling characteristic maps of a down model is done most effectively with a stride of (2, 2) and padding used to make it valid.

3.5.3 Max Pooling Layers:

The pooling layers are (2, 2) and they meet all blocks of convolution. These layers reduce longitudinality and preserve the most informative properties and improve robustness, and reduce overfitting.

3.5.4 Hierarchical Feature Learning:

Examples of low-level showcases, which are represented by the early convolutional layers include borders and contrast differences and extreme layers are trained on high-level patterns of lung tissue, which are associated with malignant and benign regions.

3.6.1 EfficientNet-B3

EfficientNetB3 is an effective convolutional neural network design that was developed to achieve high levels of precision and, additionally, has the same level of computational efficiency. It belongs to the EfficientNet family, which boasts a series of escalating process, which optimizes the depth and breadth of the network and the input resolution. It is recommended that the encouragement of more efficient representation of the data through more informative channels should use mobile inverted bottleneck convolution (MBConv) blocks with squeeze-and-innervation optimization. The architecture is particularly suitable in any health imagery task,

where the forms of shapes and textures are extremely slim and patterns play a crucial role in the diagnosis. The research adopted EfficientNetB3, which was used to create discriminative lung nodule features of CT scan images. The efficient design allowed its superior performance at reduced computation cost which was assumed to have been fixed CNN backbones.

3.6.2 Vision Transformer (ViT)

Vision Transformer (ViT) is a new deep learning model that adopts the transformer-based attention mechanism, which was first applied to natural language processing as applied to computer vision. Unlike convolutional neural networks (CNNs), which are trained to extract features using local convolutional kernels, ViT is trained to predict global contextual associations by being self-attentive in a sequence of patches of an image. ViT splits the input image into non-overlapping patches, which are of constant size. The patches are flattened and placed in a space of low dimensions. Positional embedding is an auxiliary necessity to store spatial knowledge. These patch up embeddings are further trained through multiple layers of encoders of the transformer that include multi-head self-attention and feed-forward networks. ViT uses the self-attention mechanism to detail the high-level requirements in the entire image, which is very useful in center medical imaging tasks, such as nodule category and lung cancer detection. The ViT was also utilized as a clever backbone model in this thesis, where appearance removal and grouping are used in the analysis of CT scans.

3.6.3 XGBoost Classifier

Extreme Gradient Boosting (XGBoost) is a suitable machine learning algorithm, which is founded on incline-enhanced result trees. It has enjoyed an excellent record of performance in both classification and regression, particularly when structured feature data are used. XGBoost consists of trees of decisions that are created through further training of the tree that attempts to minimize errors in the predictions of the trees that are created. This model joins the regularisation mechanisms to decrease overfitting and can be scaled at once, therefore, making it efficient and scalable. This thesis utilized XGBoost as a classifier applied on the deep features that were acquired through the usage of convoluted neural networks. In the approach, deep representation learning was employed to enhance the work of robust boosting-based decision-making in predicting lung cancer.

3.6.4 Hybrid CNN–XGBoost

To make the classification more robust and the suggested predictive accuracy more accurate, a hybrid CNN-XGBoost model was proposed in this thesis. The model is an integration of the current descent capability of deep convolutional neural networks and the high classification capability of gradient boosting. This hybrid approach uses the CNN as an automated feature extractor and is trained in the high-level representation, such as the spatial and texture-based representations on the images of lung CT scans. The deep feature vectors are taken to an XGBoost classifier rather than the ultimate classification to the entire associated layer of CNN. This combination has the benefits of both models: CNNs are more effective learning features using imaging data, and XGBoost is more successful in generalization and guided decision-making. The hybrid model was identified to be more efficient in detecting lung cancer as compared to the individual deep learning models.

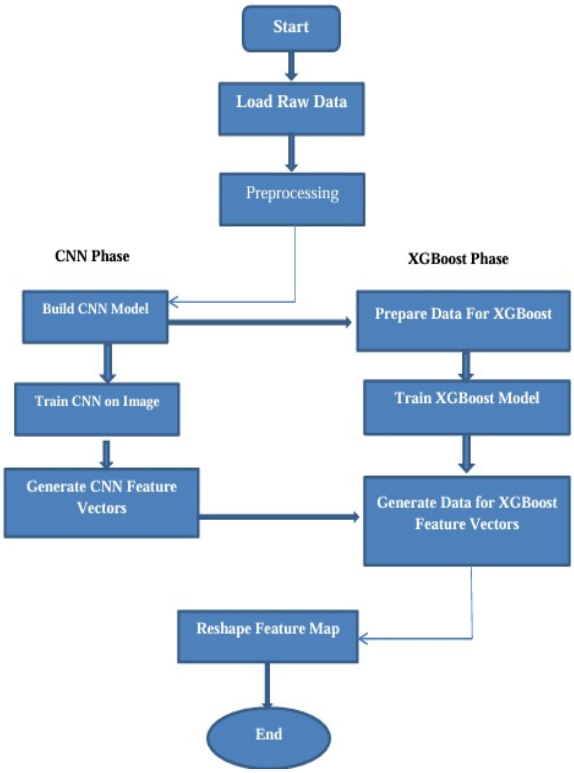

Figure: 3.6.4 CNN+XGBoost Model Flowchart

3.6.5 CNN + EfficientNetB3

The CNN + EfficientNetB3 model is a hybrid deep learning model that has been proposed as a way of enhancing a unique family in order to detect lung cancer. In this case, the EfficientNetB3 backbone is applied to complement representation learning with a convolutional neural network. EfficientNetB3 is an attribute separator that uses the pretrained model and determines complicated patterns such as nodule shape, size change, and texture of the tissue. The obtained feature maps are then input to further layers of CNN to further refine and advance the discriminative feature learning.

The two are beneficial because of the efficiency of EfficientNetB3 and the fact that CNN can acquire spatial features (hierarchy) which is highly suited to the precise classification of lung nodules and prediction of the malignancy in a CT scanner.

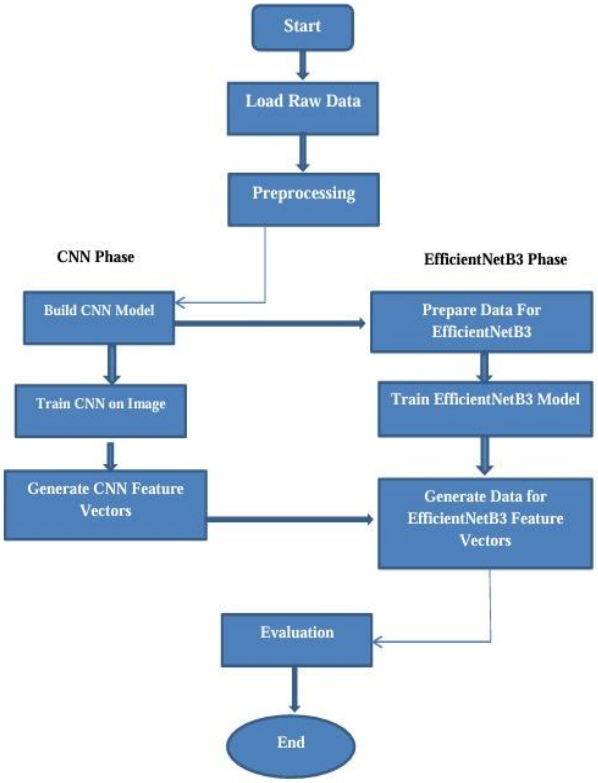

Figure: 3.6.5 CNN+EfficientnetB3 Model Flowchart

Table 3.6.5 CNN + EfficientNetB3 Model Summary

Layer (Type)	Output Shape	Number of Parameters	Connected To
Input Layer (InputLayer)	(None, 64, 64, 3)	0	–
CNN Backbone (Functional)	(None, 8, 8, 128)	93,920	Input Layer
Global Average Pooling (GlobalAveragePooling2D)	(None, 128)	0	CNN Backbone
EfficientNet-B3 (Functional)	(None, 1536)	10,783,535	Input Layer
Dense Layer	(None, 128)	16,512	Global Average Pooling
Concatenation Layer (Concatenate)	(None, 1664)	0	EfficientNet-B3, Dense Layer
Dropout Layer (Dropout)	(None, 1664)	0	Concatenation Layer
Output Dense Layer (Dense)	(None, 3)	4,995	Dropout Layer

3.6.6 NN + Vision Transformer (ViT)

The CNN + ViT network is a hybrid network that was designed in such a way that it could balance between the native feature descent and the macro appropriate understanding. CNNs can be applied in medical imaging tasks to learn fine-scale local structure and convertors can be applied in domains with long-scale structure.

In this technique, low-level spatial columns of borders, surfaces and anatomical features of lung CT images are first of all extracted with the help of a CNN. The output of these feature representations is subsequently taken into the Vision Transformer module that employs self-attention to estimate the global connection between the entire picture.

This combined architecture enhances the work of lung cancer detection since it combines the capability of CNN to learn the image spatially and the global attention mechanism of ViT. CNN+ViT is particularly an effective method in such classification problems that require local detail and global image-level information.

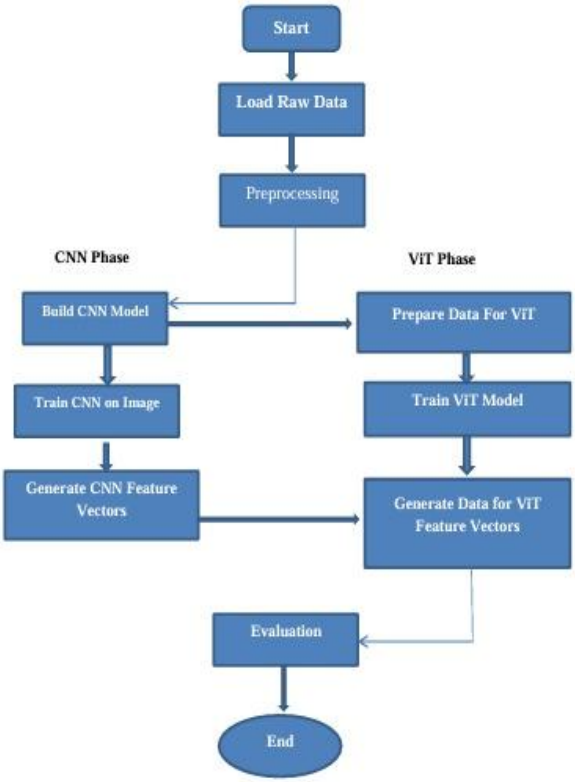

Figure: 3.6.6 CNN+ViT Model Flowchart

Table: 3.6.6: CNN + Vision Transformer (ViT) Model Summary

Layer (Type)	Output Shape	Parameters	Connected To
InputLayer	(None, 64, 64, 3)	0	-
CNN Backbone	(None, 8, 8, 128)	93,920	input_layer
Reshape	(None, 64, 128)	0	cnn_backbone
LayerNorm + Attention Blocks	(None, 64, 128)	Multiple	Transformer Encoder
GlobalAveragePooling (CNN)	(None, 128)	0	cnn_backbone
GlobalAveragePooling (ViT)	(None, 128)	0	vit_encoder
Dense Projection (CNN)	(None, 128)	16,512	cnn_gap
Dense Projection (ViT)	(None, 128)	16,512	vit_gap
Concatenate	(None, 256)	0	cnn + vit features
Dropout	(None, 256)	0	concatenate
Output Dense	(None, 3)	771	dropout

3.6.7 CNN + Vision Transformer + XGBoost

The CNN + Vision Transformer + XGBoost model can be identified as the last and most sophisticated hybrid architecture developed in this paper to increase the uncovering and risk classification effectiveness of lung cancer. It is a framework that incorporates profound spatial feature recognition, universal view knowledge and categorization on an ensemble structure.

In such architecture, low-level and mid-level spatial patterns of the lung CT and X-ray images are initially determined in the CNN part and they comprise edges, nodular features, textures and structural abnormalities. These feature maps are re-arranged and transferred to the Vision Transformer (ViT) module that makes use of multi-head self-attention tools to net long-range adjuncts and holistic background connections in the entire image. Integrating the localized chin drawing out in CNN and the global thought machines in ViT, the hybrid deep network generates fine and recognizable feature embeddings.

The last classification step is not based purely on a dense softmax layer as in the CNN+ViT model. Rather, the high-level hybrid feature embeddings by CNN +ViT backbone are provided to an Utmost Gradient Raising (XGBoost) classifier.

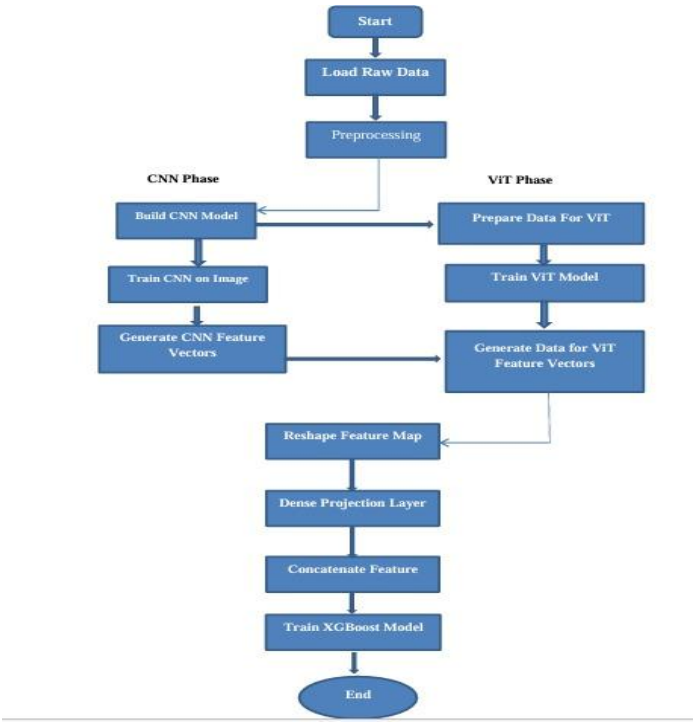

Figure:3.6.7 CNN+ViT+XGBoost Model Flowchart

Table 3.6.7 CNN +Vision Transformer +XGBoost Model Summary

Layer (Type)	Output Shape	#Params	Connected To
input_layer_7 (InputLayer)	(None, 64, 64, 3)	0	-
cnn_backbone (Functional)	(None, 8, 8, 128)	93,920	input_layer_7[0]…
reshape_3 (Reshape)	(None, 64, 128)	0	cnn_backbone[0]…
multi_head_attenti… (MultiHeadAttention…)	(None, 64, 128)	263,808	layer_normalizat…
cnn_gap (GlobalAveragePooling…)	(None, 128)	0	cnn_backbone[0]…
vit_gap (GlobalAveragePooling…)	(None, 128)	0	layer_normalizat…
dense_33 (Dense)	(None, 128)	16,512	cnn_gap[0][0]
dense_42 (Dense)	(None, 128)	16,512	vit_gap[0][0]
concatenate_3 (Concatenate)	(None, 256)	0	dense_33[0]…, dense_42[0]…
dropout_67 (Dropout)	(None, 256)	0	concatenate_3[0]…
dense_43 (Dense)	(None, 3)	771	dropout_67[0][0]

3.7 Loss Function and Optimizer:

Several deep learning and hybrid machine learning forms were also tried in this paper to recognize lung cancer and its classification using CT scan drawings. The classification task consists of three-diagnostic categories and, and thus, the problem is stated as a multi-class cataloguing mission. Therefore, one needs to select an adequate loss function and optimizer to ensure a steady convergence, accurate probability estimation, and uniform predictive ability of every architecture.

3.7.1 Loss Function for Deep Learning Models

All architectures of sources in this paper with explicit neural networks, like CNN, EfficientNetB3, Vision Transformer (ViT), CNN-EfficientNetB3, and CNN-ViT, produce class probabilities through a Softmax output layer. In this regard, these models are trained by the SCCE loss.

Mathematically, SCCE loss is written as:

L=−N1∑log(pi,yi)

where:

NNN implies the total quantity of experiments,

pi,yip_{i,y_i}pi,yi represents the calculated possibility of the true class yiy_iyi for the ithi^{th}ith sample.

3.8 Justification of Methodology

The implemented system of this research study is well organised in the sense that it provides accurate, reliable and consistent lung cancer detection and ordering of medical CT scan images. The large size of the data, which contains thousands of lung images, is necessary to enable the model to identify different variations of anatomy that correspond to benign, fatal and routine lung diseases. The data is separated into learning, confirmation, and checking divisions to ensure that sound form learning is guaranteed and reasonable performance estimation. An elaborate preprocessing pipeline is used to enhance the ability to generalize and minimize overfitting. To ensure consistency in responses and reduce the computation cost, all the pictures are resized to an equal resolution. The standards in a controlled range are made numerically stable and rapid convergence in training using the range standardization of pixel intensities. In addition to spin, horizontal flipping, and scaling, which are data expansion systems, are utilized to expand the variety of data and advocate forcefulness as opposed to imaging orientation and anatomical structures variations. Its basic methodology is based on a hybrid CNN + Vision Transformer (ViT) format. Convolutional Neural Networks will be utilized as primary feature extractors in order to achieve more delicate spatial information, including edges, textures, and patterns of pulmonary nodules. The CNNs, however, are not adequate to resolve the long-range contextual interactions in the complicated structures of lungs. To address this weakness, a Vision Transformer module is introduced that proposes multi-head self-attention operations on the image patches to enable the model to learn the global contextual relations. The resulting high-level feature embeddings of the modified CNN-ViT backbone are then forwarded to an Extreme Gradient Boosting (XGBoost) classifier. This classification step is a classification that is built on an ensemble classification that maximizes the decision boundaries, enhances separability between classes, and reduces cases of misclassification in single cases that are clinically important, like malignancies. The design methodology has been developed to offer more diagnostic strength and generalization through deep feature extraction, boosting-based structured classification.

In quantifying the proposed basis on performance valuation, the subsequent metrics are employed: Precision, Recall, F1-score, Confusion Matrix, and ROC-AUC curves, which are quantified using common medical image class measures. It is applied with the assistance of TensorFlow and Keras in terms of deep learning components, and XGBoost is used as far as the elements of the ensemble classification are concerned.

Chapter 4. Requirement Analysis, Design, and Development

4.1 Computational Requirements

The cost of the complex design of neural networks to detect lung cancer in the image of a CT scan and the image of a chest X-ray is computationally expensive due to the high dimensionality of the medical imaging data and the complex design of the hybrid neural network structure. The architectures that were tried in this paper included baseline CNN models, EfficientNetB3, ViT and networks based on a combination of them. However, in the final selected architecture, Convolutional Neural Networks (CNN), Vision Transformer (ViT), and Extreme Gradient Boosting (XGBoost) are used together in a single framework.

The CNN component can be intensively accelerated by the GPU because it contains several convolutional layers, which are accountable in the process of extracting spatial representations such as boundaries, surfaces, nodular structures, and anatomical presentations. Such processes incorporate high-dimensional matrix multiplications and repeated backpropagations, which are more costly to compute during the training process.

The ViT module is also another issue that increases the computational load, as it features a multi-head self-attention mechanism. The transformer does not focus on local receptive fields as CNNs and computes the global relationship of attention among image patches. This approach requires a larger memory and parallel processing power, particularly when handling a great number of pictures at a time.

The suggested hybrid architecture is where in the final step CNN and ViT cooperate with each other and generate high-level discriminative feature embeddings. These embeddings are then transferred to an XGBoost classifier to obtain optimized ensemble-based classification. Unlike CNN and ViT, which involve much processing on the basis of GPUs to extract the deep features, XGBoost involves the gradient boosting calculation based on CPU-based computation. The result of this extensive processing pipeline is a complicated training process, yet far more classification resistant and generalization able.

The CT data of the lungs are of huge size and quality in nature and this implies that they need efficient computation infrastructure to facilitate preprocessing, augmentation, feature extraction and iterative optimization process during the training and testing of the models. Such computation issues were the tasks solved on cloud-based systems equipped with GPUs. The overall

experimental platform was Google Colab (a project of Google) due to the fact that such platforms gave them access to NVIDIA GPU hardware, which enabled training deep learning models more quickly, and with increased efficiency in the experiment.

The CNN and the Vision Transformer modules were trained properly with the help of the GPU acceleration, and the XGBoost-based ensemble classification was backed with the help of the CPU resources. Furthermore, additional testing were conducted using domestic computing systems.

XGBoost implements the joint knowledge of boosting to improve the separability of classes as well as reducing overfitting and improving resilience to various lung conditions.

4.1.1 Environment

The expected CNN + ViT + XGBoost support was constructed and estimated using Google Colab, which may be characterized as an interactive cloud-based notebook interface that encourages systematic experimentation, displays the intermediate result, and debugs effectively. Image preprocessing pipelines, CNN feature-extractor layers, Vision Transformer attention block, and hybrid feature fusion mechanisms could be explored by trial and error in this environment.

The reason for using Google Colab was more related to the provided inbuilt GPU acceleration, which is needed during the computationally intensive deep learning process. It was applied with the help of TensorFlow and Keras to create and train CNN and Vision Transformer models, and the XGBoost library to perform gradient boosting classification using ensembles.

NumPy, OpenCV, and scientific computing libraries were also used in processing images, that was guaranteed efficient matrix operations and is reproducible. To measure the performance evaluation system, precision, exactness, memory, F1-score, and ROC-AUC measure, they were evaluated with the help of conventional machine learning evaluation instruments.

Chapter 5. Result Analysis

5.1. Confusion Matrix

One of the most crucial evaluation tools is the error matrix, and it involves the contrasting of the projected labels of class and the real labels of certainty, which gives comprehensive insight of the inscription of the classifications. The confusion matrix that would be applied to the lung CT image classification would be to establish the effectiveness of the models under test to differentiate the 3 diagnostic categories of Benign, Malignant, and Normal.

However, confusion matrices of each of the prominent architectures, such as CNN- ViT, CNN + EfficientNetB3, and CNN-XGBoost, were obtained in this paper to determine the advantages and classification errors.

CNN + EfficientNetB3 hybrid shape includes an embarrassment matrix, which is shown in figure X. The results of the predictions conducted in accordance with the most substantial predictor findings, which are the number of True Positives (TP), the number of True Negatives (TN), the number of False Positives (FP), and the number of False Negatives (FNs), characterize the correct and incorrect classification decisions, which are summarized by the matrix. This model therefore reaches the strong in the identification of theNormal lung images using 3,672 samples in the correct order and with minimal errors. A good detection rate is also attributed to the Malignant classification because it accurately predicted 708 malignant cases and this means that this classification will be very effective in the detection of cancer. The Benign one however, makes a relatively more significant error with some of the benign samples being confused with the malignant ones as there are of the same nature of the texture.

Equally, the CNN-ViT perfect misunderstanding matrix indicates that the number of classes is also adequately represented, whereas the number of classes that the CNN-ViT considers as outstanding (3,615 accurate predictions) is the highest, and the expanded global contextual representation enabled by the assistance of the transformer-based attention is also superior. However, a moderate mistake between benign and malignant variants exists since the nodules have a visual similarity at their early stage. The CNN-XGBoost models in question, the confusion matrix shows the best outcome in terms of the recognition of the largest amount of correct recognition of the Normal, the highest number of correct recognition of the Normal is 4,251. This is an indication that structured decision making would be improved in terms of deep attribute extraction and descent

advancing classification using CNN. Nevertheless, the errors of misclassification are still centralized in the. The most difficult of all the representations is the benign category.

In general, the uncertainty matrix exploration indicates that all the considered models would be extremely high in their ability to identify malignant and normal cases of lung cancer, which is extremely important in lung cancer screening. The rest of the errors are related primarily to the benign samples, as benign nodules can have a slight imaging resemblance to malignant or normal tissue structures. Such choices indicate the extent of the hybrid architectures in the propagation of the diagnostic accuracy as well as the improvement of the benign points of classification.

5.2.1 CNN + Vit + XGBoost

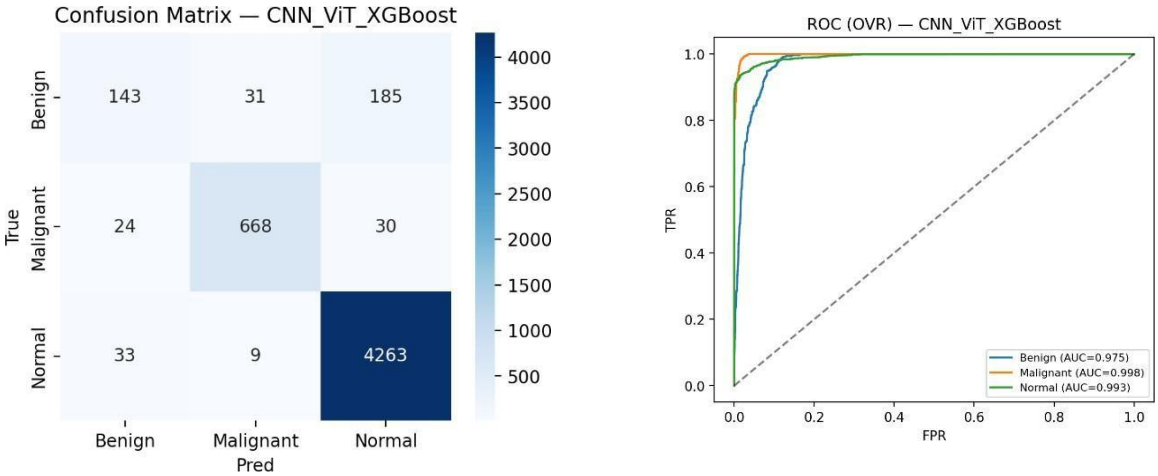

Figure: 5.2.1. CNN + Vit + XGBoost Confusion Matrix and Roc Curve

5.2.2 Confusion Matrix Summary

The CNN + Vision Transformer + XGBoost hybrid representation of the concert offered was experimented with the help of the common multi-class categorization system of measurement borrowed at the mixup matrix shown in Figure X. The model classifies the phenotype of lung CTs into three categories, that is, Benign, Malignant, and Normal.

The confusion matrix provides adequate details on accurate and inaccurate classifications into each of the classes. As per this matrix, the performance measures such as Accuracy, Precision, Remember, Specificity, and F1-score were acquired, thus, to know the performance in the general and class-specific diagnostic.

5.2.3 ROC Curve Summary

The Receiver Operating Characteristic (ROC) analysis with One-vs-Rest (OVR) approach shows that the suggested CNN + ViT + XGBoost model bears a huge discriminatory fitness in each of the diagnoses of the lung conditions. The values of Area under the Curve (AUC) of Benign, Cancerous, and Normal of 0.975, 0.998 and 0.993, respectively, were high in the model.

The almost optimal AUC of the Malignant classification denotes that the sensitivity and separability of classes are extremely high, and this is of immense benefit to clinical lung cancer detection. In the same manner, the discrimination performance is good in the Normal class. The Benign classification has a high power of classification and this is low. The general findings of the ROC have a positive effect on justifying the robustness and stability of the modelled hybrid architecture of lung cancer diagnosis in multiclass.

Table 5.2.1 Confusion Matrix Summary

True \ Predicted	Benign	Malignant	Normal	Total
Benign	143	31	185	359
Malignant	24	668	30	722
Normal	33	9	4263	4305
Total (Predicted)	200	708	4478	5386

Total Test Samples = 5,386

Correct Predictions = 143 + 668 + 4263 = 5,074

Overall Accuracy = 94.21%

Table 5.2.1 Class-wise Performance Matrix

Class	Precision	Recall (Sensitivity)	F1-Score
Benign	71.50%	39.83%	51.12%
Malignant	94.35%	92.52%	93.43%
Normal	95.20%	99.02%	97.07%

Benign Class Calculation TP = 143

FP = 24 + 33 = 57

FN = 31 + 185 = 216

Precision = TP/(TP+FP) TN = Total-(TP+FP+FN)

= 5386 − (143 + 57 + 216) = 4970

Precision

= 143 / (143 + 57)

= 143 / 200

= 71.50%

Recall

= 143 / 359

= 39.83%

Specificity

= 4970 / (4970 + 57)

= 98.87%

F1-Score

F1 = 2 × (Precision × Recall) / (Precision + Recall)

= 2 × (0.715 × 0.3983) / (0.715 + 0.3983)

= 51.12%

Malignant Class Calculation TP = 668

FP = 31 + 9 = 40

FN = 24 + 30 = 54

Precision = TP/(TP+FP) TN = Total-(TP+FP+FN)

TN = 5386 − (668 + 40 + 54) = 4624

Precision

= 668 / (668 + 40)

= 668 / 708

= 94.35%

Recall

= 668 / 722

= 92.52%

Specificity

= 4624 / (4624 + 40)

= 99.14%

F1-Score

F1 = 2 × (Precision × Recall) / (Precision + Recall)

= 2 × (94.35× 92.52) / (94.35+ 92.52)

= 93.43%

Normal Class Calculation

TP = 4263

FP = 185 + 30 = 215

FN = 33 + 9 = 42

Precision = TP/(TP+FP) TN = Total-(TP+FP+FN)

TN = 5386 − (4263 + 215 + 42) = 866

Precision

= 4263 / 4478

= 95.20%

Recall

= 4263 / 4305= 99.02%

Specificity

= 866 / (866 + 215)

= 80.11%

F1-Score

F1 = 2 × (Precision × Recall) / (Precision + Recall)

= 2 × (95.20× 99.02) / (95.20+ 99.02)

= 97.07%

5.3.1 CNN + EfficientNet-B3

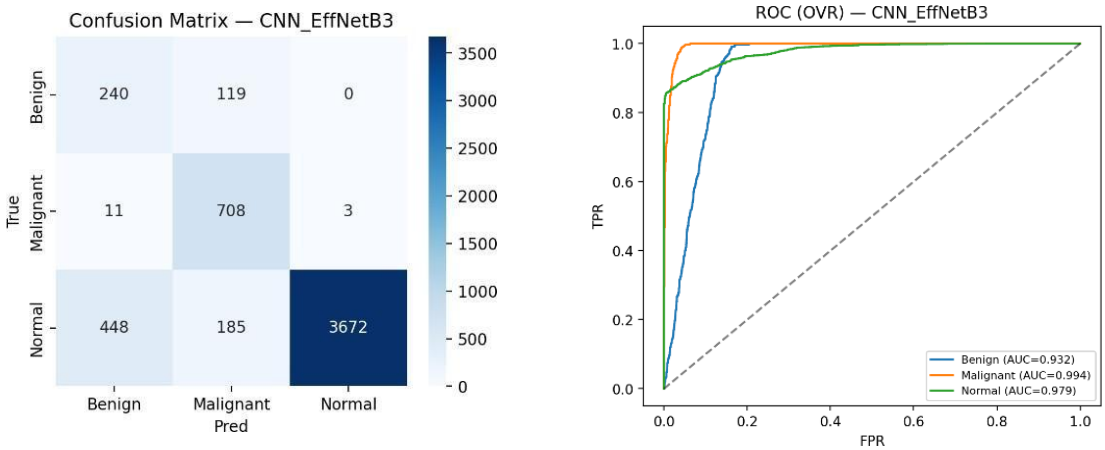

Figure : 5.2.2 CNN + EfficientNet-B3 Confusion Matrix and Roc Curve

5.3.2 Confusion Matrix Summary

The hybrid CNN + EfficientNet-B3 model was tested according to the measures of the standard multi-class classification that were received on the basis of the confusion matrix shown in the figure.

X. These measurement systems of evaluation provide the entire knowledge of the shape effectiveness to distinguish various classes of CT images of the lung. The classes of the classes that are applied in this experiment are Benign, Malignancy and Normal lung conditions. The calculated measures are correctness and accuracy, which values are measured (including the correctness and accuracy), recovery (sensitivity), and F1-score provide a reasonable concept of predictive success of both the class-based and the overall dependability of the offered crossbreed architecture.

5.3.3 ROC Curve Summary

Multiclass classification is robust in the CNN + EfficientNetB3 model proposed in the Receiver Operating Characteristic (ROC) study with the One-vs-Rest (OVR) strategy. The Area under the Curve (AUC) of respective of Gentle, Cancerous, and Regular are 0.932, 0.994 and 0.979.

The maximum AUC was in the Malignant category, and this means that it is the most separable with the highest sensitivity to detect cancer. There is also high discriminative performance in the normal class. Nonetheless, the AUC of the Benign is relatively low, and it implies that it is not a mere matter to distinguish the instances of benign and the one of other categories. The supporting answers of the CNN prove that the Total, the ROC.

+ EfficientNetB3 model also shows good performance in terms of diagnostic performance, although to a small extent lower than the hybrid CNN + ViT + XGBoost way in the future.

Table 5.3.1 Confusion Matrix Summary

Predicted	Benign	Malignant	Normal	Total
Benign	240	119	0	359
Malignant	11	708	3	722
Normal	448	185	3672	4305
Total (Predicted)	699	1012	3675	5386

Total Test Samples = 5,386

Correct Predictions = 4,620

Overall Accuracy ≈ 85.8%

Table 5.3.1 Class-wise Performance Matrix

Class	Precision	Recall (Sensitivity)	F1-Score
Benign	34.33%	66.85%	45.34%
Malignant	69.96%	98.06%	81.66%
Normal	99.92%	85.30%	92.03%

Total test samples = 5386 Benign Class Calculation

TP = 240

FP = 11 + 448 = 459

FN = 119

Precision = TP / (TP + FP) TN= total –(TP+FP+FN)

= 5386 / (240 + 459 + 119)

= 4568

Precision = TP / (TP + FP)

= 240 / (240 + 459)

= 240 / 699

= 34.33%

Recall= TP / (TP + FN)

= 240 / (240 + 119)

= 240 / 359

= 66.85%

Specificity

= 4568 / (4568 + 459)

= 90.87%

F1-Score

F1 = 2 × (Precision × Recall) / (Precision + Recall)

= 2 × 0.3433 × 0.6685 / (0.3433 + 0.6685)

= 45.34%

Malignant Class Calculation TP = 708

FP = 119 + 185 = 304

FN = 11 + 3 = 14

TN = 5386 − (708 + 304 + 14) = 4360

Precision = TP / (TP + FP)

= 708 / 1012

= 69.96%

Recall= TP / (TP + FN)

= 708 / 722 = 98.06%

Specificity = 4360 / 4664 = 93.48%

F1 = 2 × (Precision × Recall) / (Precision + Recall)

= 2 × (69.96× 93.48) / (69.96+ 93.48)

F1-Score = 81.66%

Normal Class Calculation TP = 3672

FP = 0 + 3 = 3

FN = 448 + 185 = 633

TN = 5386 − (3672 + 3 + 633) = 1078

Precision = TP / (TP + FP)

= 3672 / 3675

= 99.92%

Recall= TP / (TP + FN)

= 3672 / 4305 = 85.30%

Specificity = 1078 / 1081 = 99.72%

F1 = 2 × (Precision × Recall) / (Precision + Recall)

= 2 × (99.92× 99.72) / (99.92+ 99.72)

F1-Score = 92.03%

5.4.1 CNN + ViT

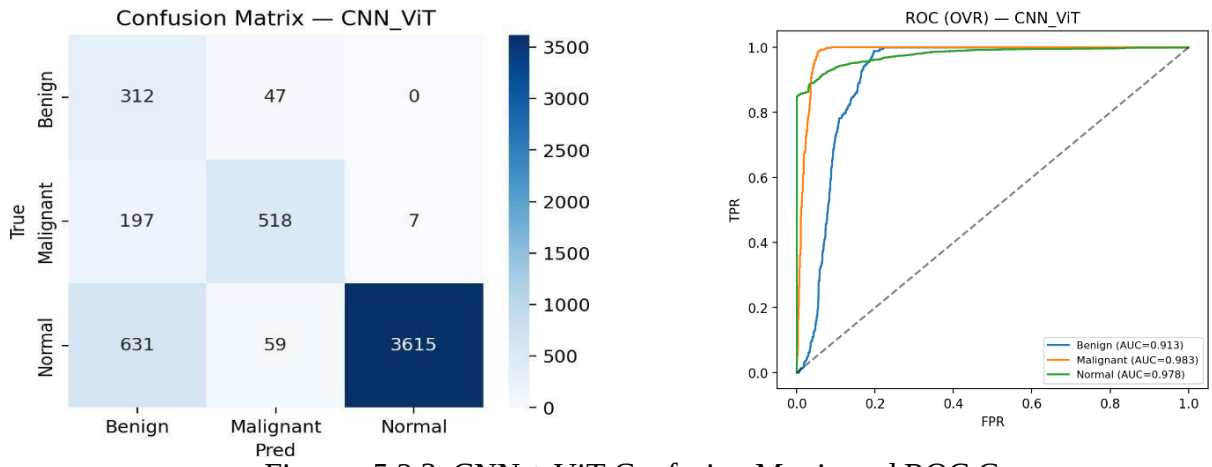

Figure : 5.2.3. CNN + ViT Confusion Matrix and ROC Curve

5.4.2 Confusion Matrix Summary

The results of the proposed hybrid CNN + Vision Transformer (CNN +ViT) model based on the uncertainty matrix of Figure X have been measured using the standard multi-class measures. These evaluation indicators provide an overview of the model with regard to the discrete lung CT image distinction. Benign, Malignant and Normal lung conditions are the classes that were tested in this experiment. As shown by the confusion matrix, the model is doing well in the recognition performance, especially in the normal class, and the capability to classify the cases of malignant and benign cases. The received numbers, including accuracy, precision, recall (sensitivity), and F1- score may comprise extensive information on how the specific class predicts and the strength and diagnostic accuracy of the chosen CNN+ViT system in general.

5.4.3 ROC Curve Summary

To achieve good multiclass discrimination implementation of the CNN + ViT standard, it is possible to monitor the Receiver Operating Characteristic (ROC) decomposition of the One-vs-Rest (OVR) method. The Area Under the Curve (AUC) of the Benign, Malignant and normal has values of 0.913, 0.983 and 0.978, respectively.

The AUC was maximum in the malignant class and it implies that it is very sensitive and can easily distinguish between other classes. The discriminative power is also huge in the Normal class. The AUC of the benign class is, however, smaller; this is because it was not large enough to separate benign cases and malign and normal cases. All in all, the results of ROC show that CNN + ViT architecture is a practical classification model with fewer results compared to the ultimate hybrid CNN + ViT + XGBoost model.

Table 5.4.1 Confusion Matrix Summary Table

True \ Predicted	Benign	Malignant	Normal	Total
Benign	312	47	0	359
Malignant	197	518	7	722
Normal	631	59	3615	4305
Total	1140	624	3622	5386

Total Test Samples = 5,386

Correct Predictions = 4,445

Overall Accuracy ≈ 82.5%

Table 5.4.1 Class-wise Performance Matrix

Class	Precision	Recall (Sensitivity)	F1-Score
Benign	0.27	0.87	0.41
Malignant	0.83	0.72	0.77
Normal	0.998	0.84	0.91

Benign Class Calculation TP = 312

FP = 828

FN = 47

Precision = TP / (TP + FP)

= 312 / (312+828)

Precision = 0.274

Recall= TP / (TP + FN)

= 312 / (312+47)

Recall = 0.869

F1 = 2 × (Precision × Recall) / (Precision + Recall)

= 2 × (0.274 × 0.869) / (0.274 + 0.869)

= 2 × 0.238 / 1.143

F1 ≈ 0.416

Malignant Class Calculation TP = 518

FP = 106

FN = 204

Precision = TP / (TP + FP)

= 518 / (518+106)

Precision = 0.830

Recall = TP / (TP + FN)

= 518 / (518+204)

Recall = 0.717

F1 = 2 × (Precision × Recall) / (Precision + Recall) 2 × (0.830 × 0.717) / (0.830 + 0.717)

F1-score ≈ 0.769

Normal Class Calculation TP = 3615

FP = 7

FN = 690

Precision = TP / (TP + FP)

= 3615 / (3615+7)

Precision = 0.998

Recall = TP / (TP + FN)

= 3615 / (3615+690)

Recall = 0.840

F1 = 2 × (Precision × Recall) / (Precision + Recall) 2 × (0.998× 0.840) / (0.998+ 0.840)

F1-score ≈ 0.913

5.5.1CNN + XGBOOST

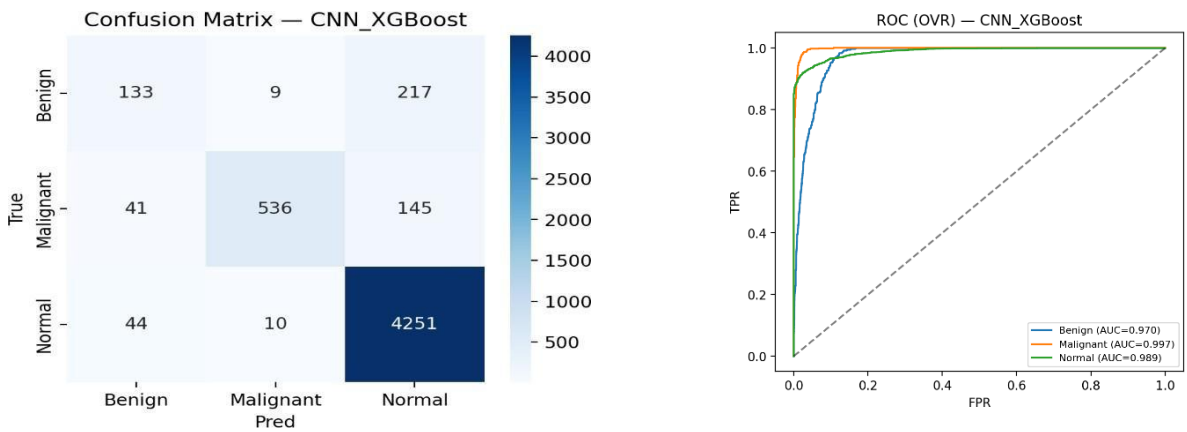

Figure: 5.2.4. CNN + XGBOOST Confusion Matrix and Roc Curve

5.5.2 Confusion Matrix Summary

The mix-up matrix provides the performance of the future CNN+XGBoost hybrid ensemble on the three classes of lung conditions: Benign, Malignant, and Normal.

It has proven that the model has been efficient in predictive capabilities, which is the identification of normal and malignant cases. The fact that most of the normal samples are rightfully identified implies that a good identification of the normal patterns of the lung is made. On the same note, the malignant category possesses a high correct detection and the image indicates the potential of the model to detect clinically significant cancerous nodules at reduced false negative. Despite the fact that there are some specific cases of benignity identified as normal, the overall confusion matrix validates the fact that the CNN+XGBoost model is far stronger in nature. The diagnostic reliability of such an architecture is much higher as compared to other architectures that are considered. Therefore, it is a better choice to be used and be the final LungNet-RiskX architecture.

5.5.3 ROC Curve Summary

The CNN + XGBoost model in the operating strategy under One-vs-Rest (OVR) is shown to be working as a multiclass classification in the operating Characteristic of the Receiver Operating Characteristic (ROC). The Benign, Malignant and normal Area Under the Curve (AUC) are 0.970, 0.997, and 0.989, respectively.

The highest AUC signifying good sensitivity and almost perfect ability to separate was realized in the Malignant class, which is clinically pertinent in the determination of cancer, devoid of errors. The Normal type is also highly separable and the AUC value of this type is also high. However, the Benign, being a little below the other classes, is good in classification performance.

Table 5.5.1 CNN + XGBOOST Confusion Matrix

True \ Predicted	Benign	Malignant	Normal	Total
Benign	133	9	217	359
Malignant	41	536	145	722
Normal	44	10	4251	4305
Total	218	555	4613	5386

Total Test Samples = 5,386

Correct Predictions = 4,920

Overall Accuracy ≈ 91.3%

Table 5.5.1 CNN + XGBOOST Class-wise Performance Matrix

Class	Precision	Recall	F1-score
Benign	0.610	0.370	0.460
Malignant	0.966	0.742	0.839
Normal	0.922	0.988	0.954
Overall Accuracy	-	-	91.3%

Benign Class Matrix TP = 133
FP = 85
FN = 226

Precision = TP / (TP + FP)
= 133 / (133 + 85)
= 133 / 218
Precision ≈ 0.610

Recall = TP / (TP + FN)
= 133 / (133 + 226)
= 133 / 359
Recall ≈ 0.370

F1 = 2 × (Precision × Recall) / (Precision + Recall)
= 2 × (0.610× 0.370) / (0.610+ 0.370)
F1-score ≈ 0.460 Malignant Class Matrix TP = 536
FP = 19
FN = 186

Precision = TP / (TP + FP)
= 536 / (536 + 19)
= 536 / 555

Precision ≈ 0.966 Recall = TP / (TP + FN)
= 536 / (536 + 186)
= 536 / 722
Recall ≈ 0.742

F1 = 2 × (Precision × Recall) / (Precision + Recall)
= 2 × (0.966× 0.742) / (0.966+ 0.742)
F1-score ≈ 0.839

Normal Class Matrix TP = 4251
FP = 362
FN = 54

Precision = TP / (TP + FP)
= 4251 / (4251 + 362)
= 4251 / 4613
Precision ≈ 0.922

Recall = TP / (TP + FN)
= 4251 / (4251 + 54)
= 4251 / 4305
Recall ≈ 0.988

F1 = 2 × (Precision × Recall) / (Precision + Recall)
= 2 × (0.922× 0.988) / (0.922+ 0.988)
F1-score ≈ 0.954p

Correct Predictions = 4920
Total Test Samples = 5386
Overall Accuracy = 4920 / 5386 ≈ 91.3%

5.6 Accuracy Comparison

The results of the validation accuracy provide a comparative impression of the classification capability of the examined models in detail. The CNN+ViT) The CNN models has a validation of 82.55, which indicates that it is efficient in terms of learning that is based on convolutional quality descent and transformer-based appropriate modelling.

The model was concluded to have some shortcomings in the confidence of finedifference between the various types of multiclass lung CT and yet it was yielding realistic results.

CNN + EfficientNetB3 model reported the best validation accuracy of 85, which was slightly higher than CNN + ViT configuration. This is due to the fact that EfficientNetB3 is scaled with the most efficient compound, thereby making its extraction of features more efficient and extrapolating better on unknown data.

The CNN + XGBoost mixed model, followed by a validation truth of 90 percent, was an indicator of the utility of the use of deep CNN-extracted features combined with gradient boosting-based classification. This type of combination enhanced the discriminative learning rate and decreased the misclassification, especially of the clinically significant cases of malignancy.

However, most importantly, the proposed CNN + ViT + XGBoost (Final Model) obtained a maximum confirming accuracy of 94.21, which was even higher than those of other designs studied. The high multiclass diagnostic power and the generalization is also increased with such an upgrade enhancement, indicating that the hybrid feature fusion is also efficient besides the fact that the boosting based optimization of the decision making increases the overall generalization. It is possible to conclude that CNN+ViT+XGBoost is the most stable model, the best architecture in the LungNet-RiskX system.

Table 5.6 Performance Comparison

Model Architecture	Validation Accuracy (%)	Benign AUC	Malignant AUC	Normal AUC
CNN + Vision Transformer (CNN+ViT)	82.55%	0.926	0.984	0.972
CNN + EfficientNetB3	85%	0.940	0.980	0.970
CNN + XGBoost	90%	0.961	0.996	0.985
CNN + ViT + XGBoost (Final Model)	94.21%	0.975	0.998	0.93

5.7 Compare Matrix

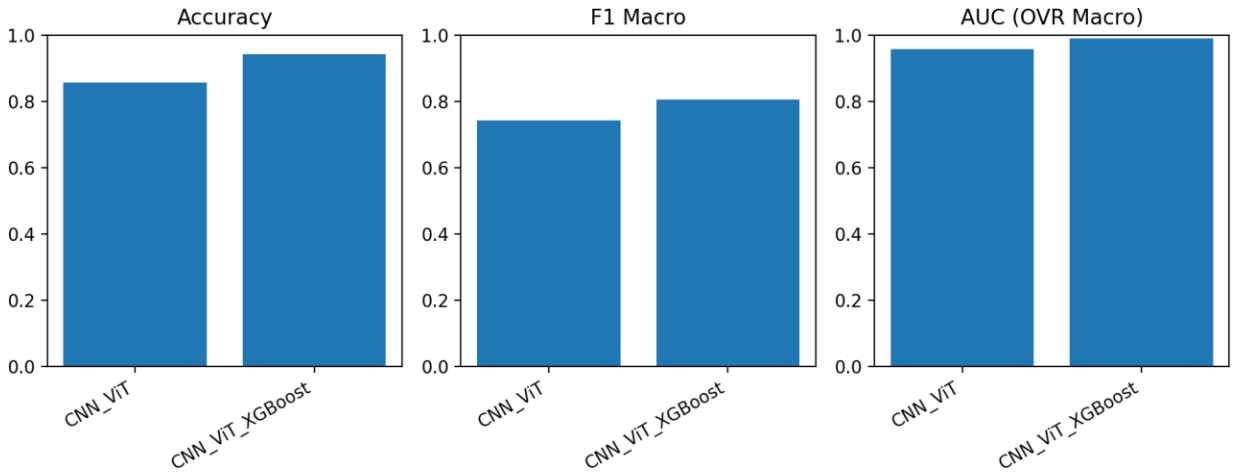

Figure:5.7 Compare Matrix

The metric comparison reveals that the CNNViT representation in which the CNNViT is hybridly glued has better results in comparison with all the evaluation metrics compared to the CNNViT standalone representation. The accuracy of CNNViTXGBoost is approximately 0.94, which is rather high as compared to CNNViT (approximately 0.85). On the same note, F1 Macro score increases by an average of 0.740.80, which would suggest improved performance of the classes. According to the case of AUC (OVR Macro), both models have a high level of discriminatory power, with CNNViTXGBoost having the highest (approximately 0.99) and the second highest (approximately 0.96) discriminatory power as compared to the other two models. Altogether, there is a better classification run of the hybrid CNNViTXGBoost model.

5.8 Loss and Training Analysis

CNN + EfficientNetB3 Training Analysis

To estimate the performance of the CNN + EfficientNetB3 hybrid model during training, the change of the loss and accuracy curve was estimated in terms of age. Figure X showed that the learning process involved the process of educating and lost confirmation that declined gradually and the accuracy gained slowly.

The loss curves are converged smoothly, which shows that features are optimized and learnt successfully. On the same note, training accuracy and validation accuracy were gradually improving to approximately 0.83 training accuracy and 0.85 validation accuracy, which depicts a high generalization ability.

Furthermore, the orientation of the folds among the support curve and the exercising curve presupposes minimal overfitting and, therefore, ensures that the model is beneficial in acquiring meaningful patterns of lung CT images, and not sensitive to unfamiliar data. Overall, the convergence behavior testifies to the feasibility of EfficientNetB3 as a handy schoolboy of the feature extraction in risk prediction of lung cancer.

CNN + Vision Transformer (ViT) Training Analysis

It was found that the accuracy and loss patterns of the anticipated model of the hybrid CNN + Vision Transformer (CNN+ViT) were monitored to determine the training performance of the model. Figure X demonstrates that, equally exercising and confirmation loss, and classification accuracy were gradually decreasing and constantly increasing, respectively in the show.

The loss curves suggest that the convergence of the hybrid architecture should be consistent, which proves that the hybrid architecture can effectively train the local spatial representations with the help of CNN layers and the global contextual representations with the help of transformer-based self-attention. The accuracy of the exercise and authentication was slowly growing with approximations of 0.82 exercise and 0.83 validation accuracy.

Moreover, it is also indicated by the fact that the slight disconnect between the exercise curve and the validation curve means that one is likely to experience an overfitting that is supervised and good performance during the generalization process. This may be justified by the fact that the application of CNN features and ViT attention mechanisms improves the diagnostic performance of the multiclass lung cancer diagnostics.

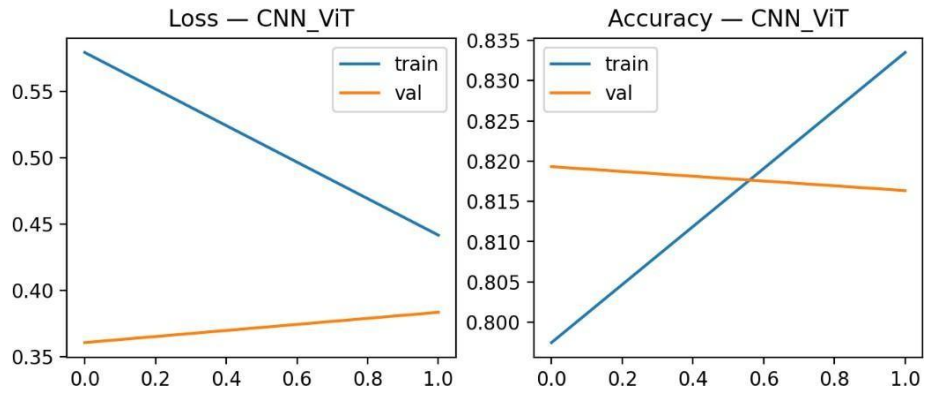

Figure: 5.8 Loss and Training Analysis

CNN + XGBoost Performance Analysis

The CNN + XGBoost hybrid model will be trained in two stages, unlike a deep learning architecture. Under this approach, CNN as a deep feature extractor, and XGBoost as the eventual organiser, are involved, which considers the extracted feature vectors.

This implies that this hybrid pipeline is more accurate in prediction than the other models, as shown by the analysis of the performance. With the help of the CNN-extracted deep representations, the XGBoost can make the difference between benign, malignant, and normal classes with the ultimate accuracy of about 0.91, as Figure X reveals.

The assumption is that the joint use of CNN-based automatic feature learning and the ability of a structured decision-making of the gradient boosting has its benefits that are demonstrated in this performance. Another factor is that the CNN+XGBoost model has better credibility and low rates of misclassification, especially in the clinically severe cases of malignancy and this is why this model is the most credible as part of the LungNet-RiskX model.

CNN + XGBoost Performance Analysis

The CNN + XGBoost hybrid model will be trained in two stages, unlike a deep learning architecture. CNN may be used in this method to obtain the deep features and to classify the final results; XGBoost may be used based on the extracted feature vectors.

This implies that this hybrid pipeline is more accurate in prediction than the other models, as shown by the analysis of the performance. With the help of the CNN-extracted deep representations, the

XGBoost can make the difference between benign, malignant and normal classes with the ultimate accuracy of about 0.91, as Figure X reveals.

The assumption is that the joint use of CNN-based automatic feature learning and the ability of a structured decision-making of the gradient boosting has its benefits that are demonstrated in this performance. Another factor is that the CNN+XGBoost model has better credibility and low rates of misclassification, especially in the clinically severe cases of malignancy, and this is why this model is the most credible as part of the LungNet-RiskX model.

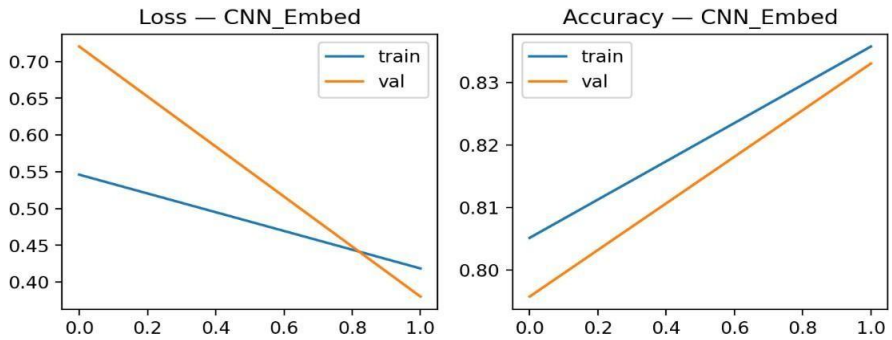

Figure: 5.8 Loss and Training Analysis

5.9 Train Time Comparison

The comparative analysis on the training time shows that the number of differences in the computational efficiency between the architecture under consideration is enormous. The CNNViT model is the most complicated model to compute, since in the first comparison, this model took the longest time to be trained, specifically, it took around 15,000 seconds to be trained. The CNNEfficientNetB3 was more effective, and the time taken on the training model was close to 3,000 seconds. CNNXGBoost model, in its turn, took only several hundred seconds, which suggests that it should have a significantly lower set of computational requirements.

In the second comparison, CNNViT was estimated to take about 8000 seconds in contrast to the hybrid CNNViTXGBoost model, which took a few hundred seconds to train the model. Such results indicate that the efficiency of the computations does not decline in the case of XGBoost, which has resulted in a drop in the efficiency of the classification.

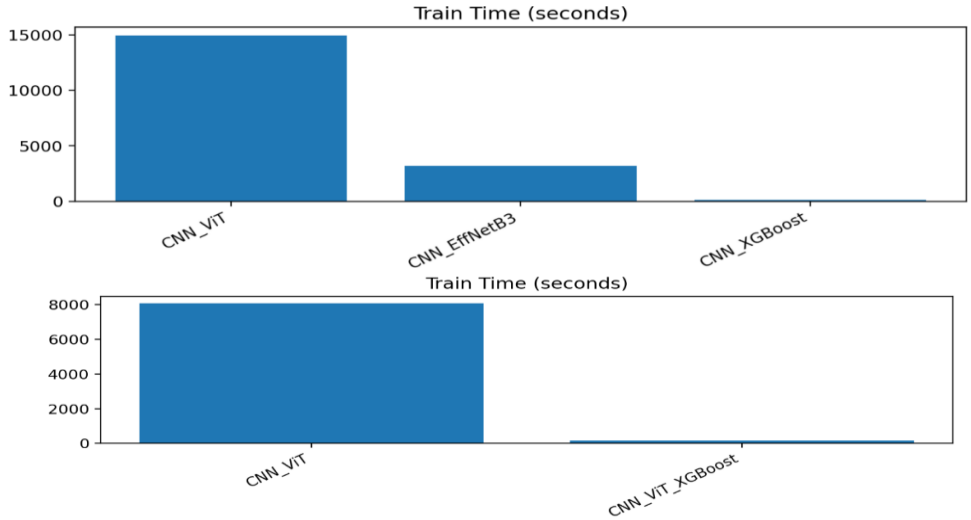

Figure 5.9 Training Time Comparison Models

5.10 Risk Prediction and Probability Analysis

5.10.1 Output-Based Risk Probability Interpretation

The hybrid CNN-ViT-XGBoost algorithm is proposed, which is an algorithm producing the class-wise probability performance per-image score of each lung CT, and the performance is not categorical. Such a probabilistic form of representation is more explicatory and can be employed in the provision of risk-based clinical assessment.

The probability estimates, which were modeled, in the case of the illustrated sample case, were as follows: Benign: 0.5644.

Malignant: 0.0737

Normal: 0.3619

According to the maximum probability case (56.44%), this model considered the image to be Benign.

The noninvasive type that the model showed more opportunities for is irrespective of the fact that the file name is a signal of a normal case. It is noteworthy that the risk of malignancy is very minimized (7.37 percentage), and this fact assumes that the occurrence of this sample does not expose them to the risk of developing cancer too frequently.

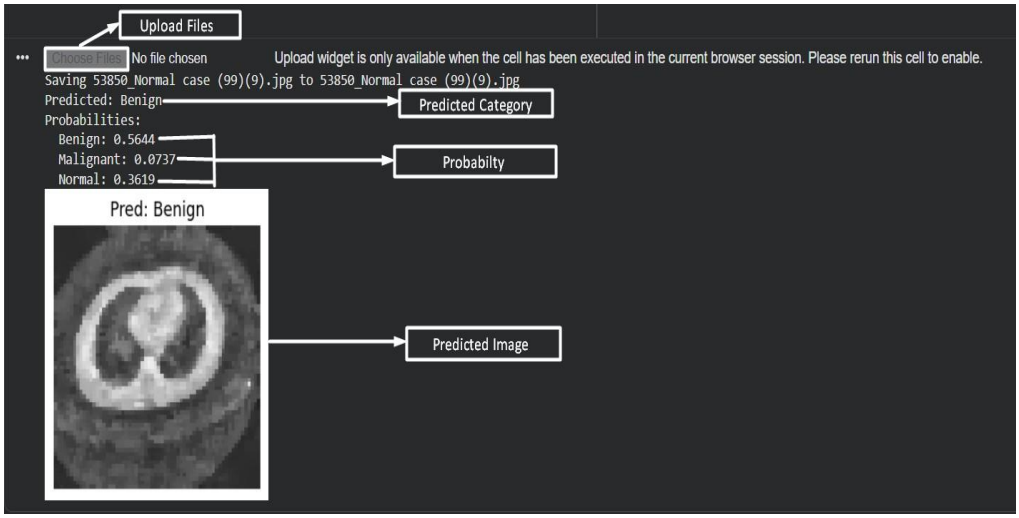

Figure 5.10.1 Sample Risk Prediction Output for a Lung CT Image

5.10.2 Risk Confidence and Decision Reliability:

The probability distribution demonstrates several important characteristics:

1. Medium level of confidence in the benign type.
2. Second-level probability distribution to the normal class.
3. Delays cancerous potentiality to a great extent.

This distribution demonstrates that the malignant characteristics are far apart to be separated by the model, with a bit of interference from the benign and normal populations. Such overlap is bound to happen because of similarities in the structure of the non-cancerous lung tissues.

The bad malignant confidence, in this instance, is high in the pattern of discrimination of cancerous images as compared to non-cancerous images.

5.10.3 Clinical Risk Interpretation:

The clinical indication that the sampled group is at low risk of being exposed to oncology risks is the low malignancy (<10%).

The Malignancy suspicion is not a strong entity that is strong and the type of classification is only

benign and normal. The system of the specified case in this connection possesses uniform malignant risk subduction.

5.10.4 Significance of Probability-Based Risk Modeling:

The incorporation of probability-based outputs significantly enhances the clinical utility of LungNet-RiskX.

Instead of generating a hard-label decision alone, the system provides:

1. Quantified prediction confidence
2. Transparent risk distribution
3. Improved interpretability for clinical assessment

These probabilistic models are better suited to the real-world process of making diagnoses as compared to traditional categorical-only models.

5.10.5 Contribution of the Risk Module:

The risk prediction module:

1. Produces intelligible probability scores.
2. Great levels of highly malignant discrimination.
3. Bias to confidence-based clinical reasoning.
4. Improves the disclosure of diagnosis.

Chapter 6. Conclusions

The proposal (LungNet-RiskX) presented in the paper is a hybrid deep learning structure, the LungNet-RiskX, programmable lung cancer detection and multiclassification and risk assessment on the basis of the CT scan and a chest X-ray image data provided of the patient in question. The created system combines Convolutional Neural Networks (CNN), Vision Transformers (ViT), and Extreme Gradient Boosting (XGBoost) in order to enhance the degree of diagnostic quality and model accuracy.

A preprocessing strategy was utilized which included the following steps: image resizing, pixel normalization, contrast enhancement using CLAHE, noise elimination, and data augmentation to be able to standardize the dataset and make the most of generalization. The CNN-ViT combined structure allowed high-level features to be learnt by explaining the small-scale local features and the large-scale contextual relationship of lung images. These high-level feature representations were then input into the XGBoost classifier that allowed for further precision in class differentiation and reduced the amount of misclassification errors.

The CNN was evaluated to be doing well by estimating the performance on a dataset of 53,853 lung images.

The hybrid model + ViT + XGBoost became more successful in comparison with other setups. This final configuration resulted in bringing the overall accuracy of 94.21 and good ROC-AUC scores of all the diagnostic categories and excellent discriminative performance of the cancerous cases.The interpretation of the model was also improved through the risk output (in the form of probabilities) because the predictions became of the nature of confidence sensitivity. It is a probabilistic model, which helps to make clinical decisions obtained and improve the clarity of the diagnostic process.

6.1 Future Work

The following section of our study will be dedicated to the following stage of the LungNet-RiskX system implementation in the actual clinical practice. This will involve the design of a web and mobile-based application that will support real-time usage of the hospitals. Besides this, the friendly interface will be proposed, which will enable inputting the corresponding information about the patient, i.e., age, smoking history, and identifying the contributing factors, as this will help assess the risk and improve the workability of the system overall.

Reference:

Abdullahi, K., Kannan, R. and Ali, A.B., 2025. Deep learning techniques for lung cancer diagnosis with computed tomography imaging: A systematic review for detection, segmentation, and classification. Information, 16(6), p.451. https://doi.org/10.3390/info16060451

Al-Ameer, A.A.A., Hussien, G.A. and Ameri, H.A.A., 2022. Lung cancer detection using image processing and deep learning. Indonesian Journal of Electrical Engineering and Computer Science, 28(2), p.987. https://doi.org/10.11591/ijeecs.v28.i2.pp987-993

Bushara, A.R., Vinod Kumar, R.S. and Shahi, D., 2022. Deep learning-based lung cancer classification of CT images using augmented convolutional neural networks. Electronic Letters on Computer Vision and Image Analysis, 21(1), pp.130–142.

Chai, C.C., Khoh, W.H., Pang, Y.H. and Yap, H.Y., 2024. A lung cancer detection with pre- trained CNN models. Journal of Informatics and Web Engineering, 3(1), pp.41–54. https://doi.org/10.33093/jiwe.2024.3.1.3

Faizi, M.K. et al., 2025. Deep learning-based lung cancer classification of CT images. BMC Cancer, 25(1), p.1056. https://doi.org/10.1186/s12885-025-14320-8

G, A.I.S. et al., 2015. Data from LIDC-IDRI. The Cancer Imaging Archive. https://doi.org/10.7937/k9/tcia.2015.lo9ql9sx

Huang, K.-Y., Chung, C.-L. and Xu, J.-L., 2025. Deep learning object detection-based early detection of lung cancer. Frontiers in Medicine, 12, p.1567119. https://doi.org/10.3389/fmed.2025.1567119

Lee, J.H. et al., 2020. Performance of a deep learning algorithm compared with radiologic interpretation for lung cancer detection on chest radiographs in a health screening population. Radiology, 297(3), pp.687–696. https://doi.org/10.1148/radiol.2020201240

Makaju, S., Prasad, P.W.C., Alsadoon, A., Singh, A.K. and Elchouemi, A., 2018. Lung cancer detection using CT scan images. Procedia Computer Science, 125, pp.107–114. https://doi.org/10.1016/j.procs.2017.12.016

Meeradevi, T. et al., 2025. Lung cancer detection with machine learning classifiers with multi- attribute decision-making system and deep learning model. Scientific Reports, 15(1), p.8565. https://doi.org/10.1038/s41598-025-88188-w

Ozdemir, O., Russell, R.L. and Berlin, A.A., 2019. A 3D probabilistic deep learning system for detection and diagnosis of lung cancer using low-dose CT scans. IEEE Transactions on Medical Imaging, 39(5), pp.1419–1429. https://doi.org/10.1109/tmi.2019.2947595

Patel, S., Gawande, U. and Singh, A., 2019. Challenges in lung nodule detection and classification in CT images. Journal of Medical Imaging, 6(2), pp.1–12.

Ponnada, V.T. and Srinivasu, S.V.N., 2019. Efficient CNN for lung cancer detection. International Journal of Recent Technology and Engineering (IJRTE), 8(2), pp.3499–3503. https://doi.org/10.35940/ijrte.b2921.078219

Raut, S., Patil, S. and Shelke, G., 2021. Lung cancer detection using machine learning approach. International Journal of Advance Scientific Research and Engineering Trends, 6(1), pp.47–51. https://doi.org/10.51319/2456-0774.2021.0005

Shariff, V. et al., 2025. Optimizing non small cell lung cancer detection with convolutional neural networks and differential augmentation. Scientific Reports, 15, p.15640. https://doi.org/10.1038/s41598-025-98731-4

Simonyan, K. and Zisserman, A., 2014. Very deep convolutional networks for large-scale image recognition. arXiv preprint arXiv:1409.1556.

World Health Organization, 2021. Cancer fact sheet: Lung cancer statistics. Available at: https://www.who.int/ (Accessed: Day Month Year).

Downloading TCIA Images – TCIA Online Help – Cancer Imaging Archive Wiki, no date. Availableat:https://wiki.cancerimagingarchive.net/display/NBIA/Downloading+TCIA+Images